Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2.

Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2.

Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2. For the production of x_1 units of P_1 and x_2 units of P_2, you're profit in € is:

$$G(x_1, x_2) = 30x_1 + 50x_2$$

Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2. For the production of x_1 units of P_1 and x_2 units of P_2, you're profit in € is:

$$G(x_1, x_2) = 30x_1 + 50x_2$$

Three machines M_A, M_B and M_C produce the required components A, B and C for the products.

M_A :
M_B :
M_C :

Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2. For the production of x_1 units of P_1 and x_2 units of P_2, you're profit in € is:

$$G(x_1, x_2) = 30x_1 + 50x_2$$

Three machines M_A, M_B and M_C produce the required components A, B and C for the products. The components are used in different quantities for the products

$$\begin{aligned} M_A: &\quad 4x_1 + 11x_2 \\ M_B: &\quad x_1 + x_2 \\ M_C: &\quad x_2 \end{aligned}$$

Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2. For the production of x_1 units of P_1 and x_2 units of P_2, you're profit in € is:

$$G(x_1, x_2) = 30x_1 + 50x_2$$

Three machines M_A, M_B and M_C produce the required components A, B and C for the products. The components are used in different quantities for the products, and each machine requires some time for the production.

$$\begin{aligned} M_A: &\quad 4x_1 + 11x_2 \leq 880 \\ M_B: &\quad x_1 + x_2 \leq 150 \\ M_C: &\quad x_2 \leq 60 \end{aligned}$$

Maximizing Profits

You're the boss of a small company that produces two products P_1 and P_2. For the production of x_1 units of P_1 and x_2 units of P_2, you're profit in € is:

$$G(x_1, x_2) = 30x_1 + 50x_2$$

Three machines M_A, M_B and M_C produce the required components A, B and C for the products. The components are used in different quantities for the products, and each machine requires some time for the production.

$$\begin{aligned} M_A: &\quad 4x_1 + 11x_2 \leq 880 \\ M_B: &\quad x_1 + x_2 \leq 150 \\ M_C: &\quad x_2 \leq 60 \end{aligned}$$

Which choice of (x_1, x_2) maximizes the profit?

Solution

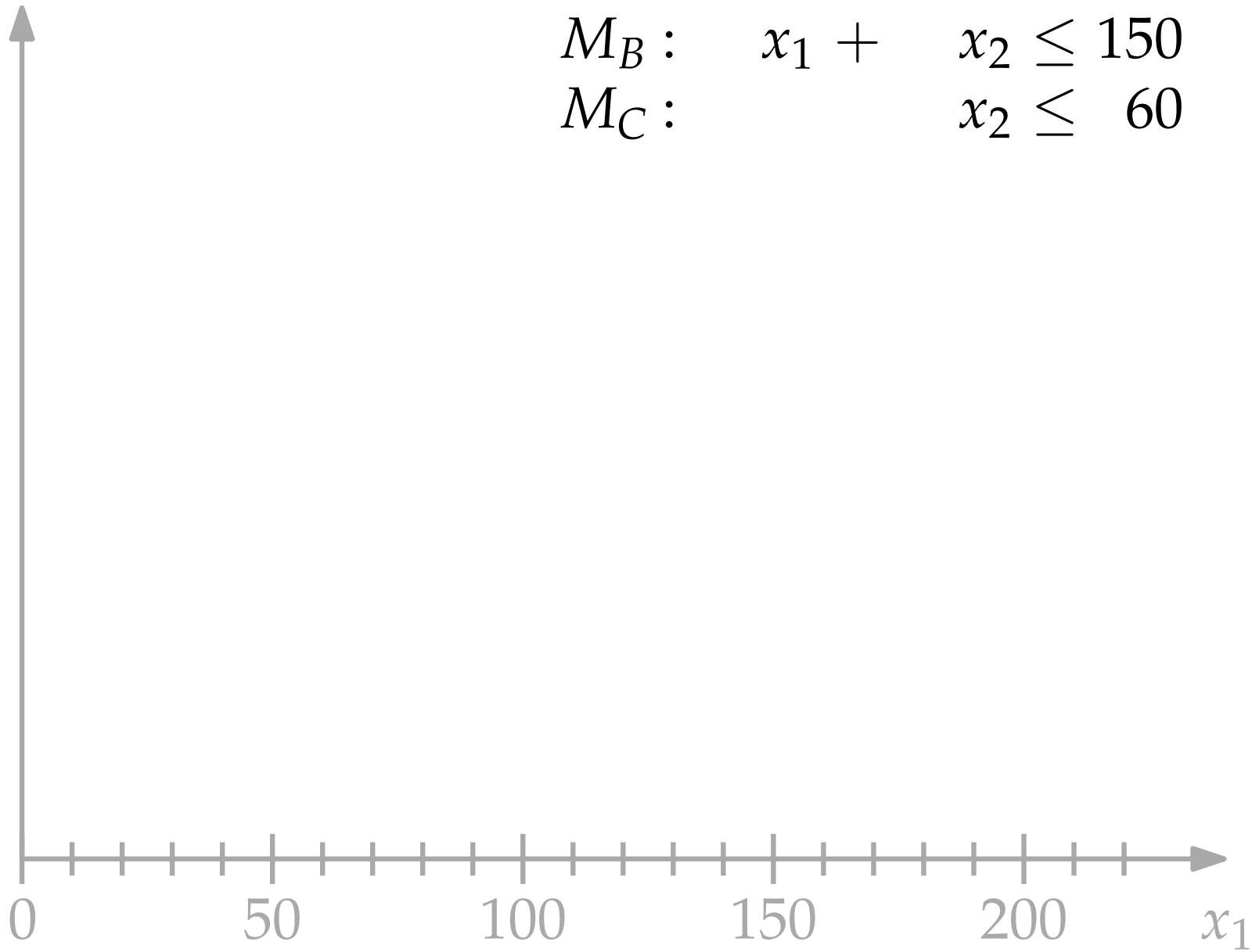
Linear constraints:

$$
\begin{aligned}
M_A: &\quad 4x_1 + 11x_2 \leq 880 \\
M_B: &\quad x_1 + x_2 \leq 150 \\
M_C: &\quad x_2 \leq 60
\end{aligned}
$$

Solution

Linear constraints:

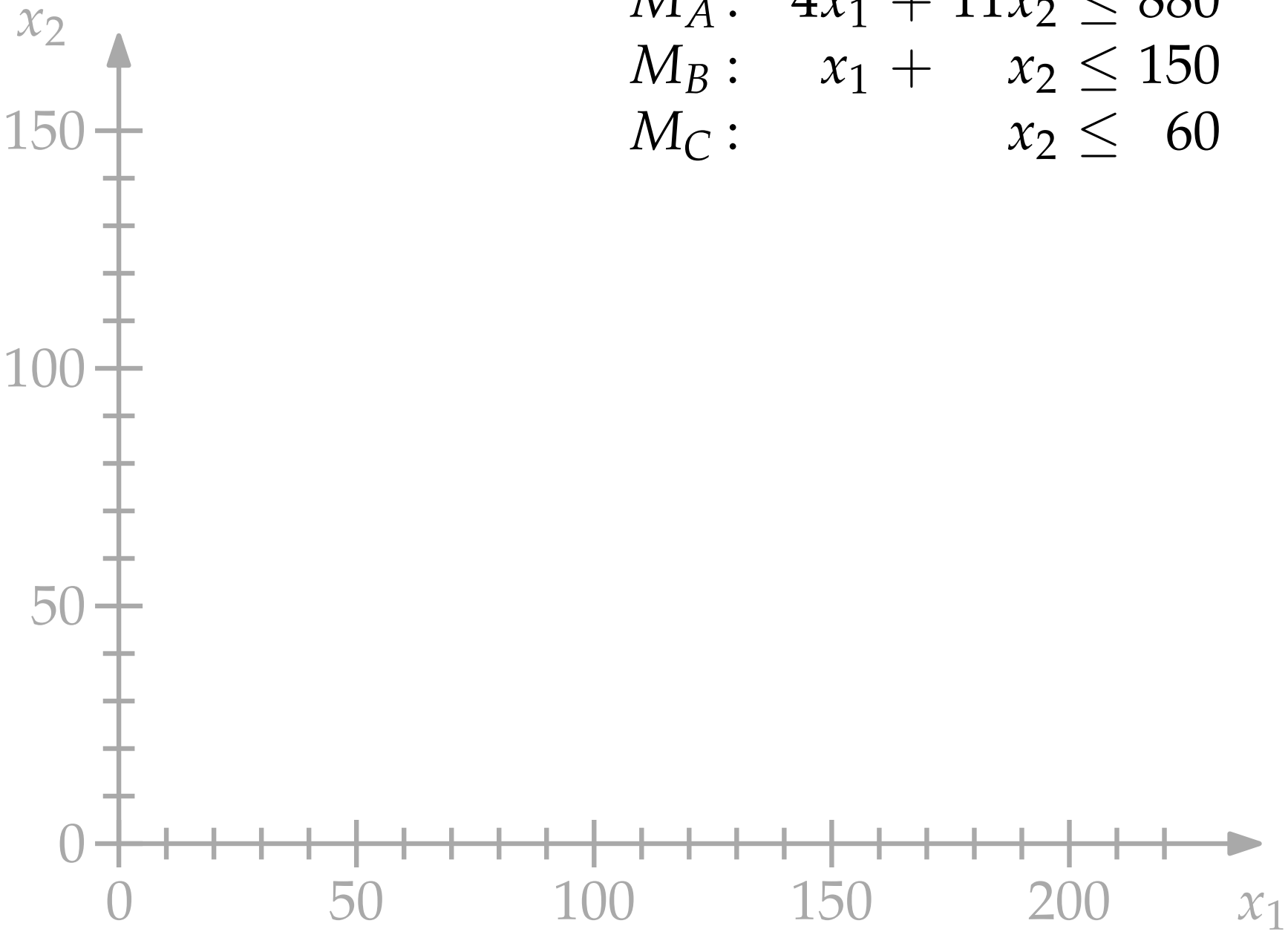
$$
\begin{aligned}
M_A: &\quad 4x_1 + 11x_2 \leq 880 \\
M_B: &\quad x_1 + x_2 \leq 150 \\
M_C: &\quad x_2 \leq 60
\end{aligned}
$$



Solution

Linear constraints:

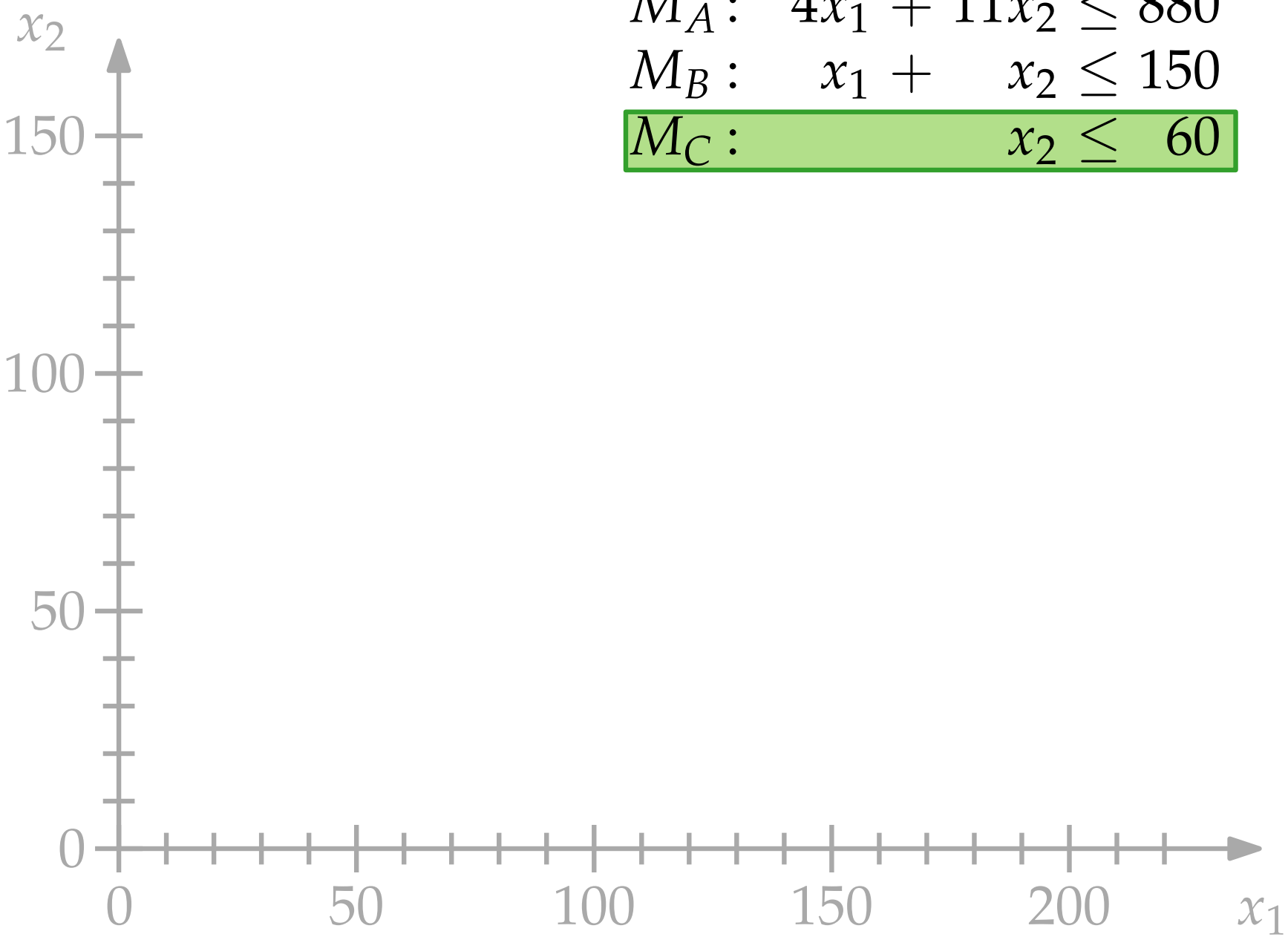
$$
\begin{aligned}
M_A: &\quad 4x_1 + 11x_2 \leq 880 \\
M_B: &\quad x_1 + x_2 \leq 150 \\
M_C: &\quad x_2 \leq 60
\end{aligned}
$$



Solution

Linear constraints:

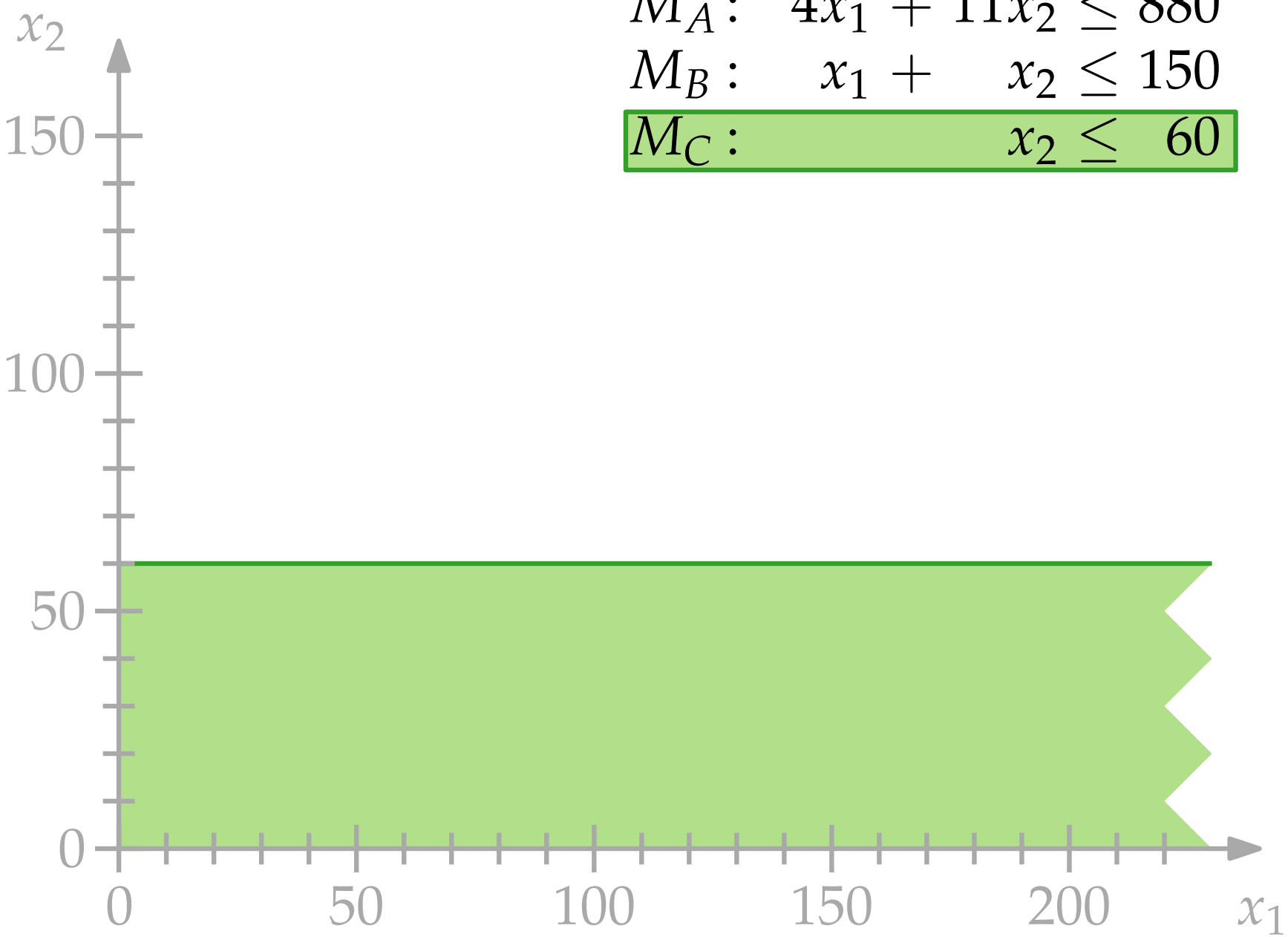
$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$



Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$



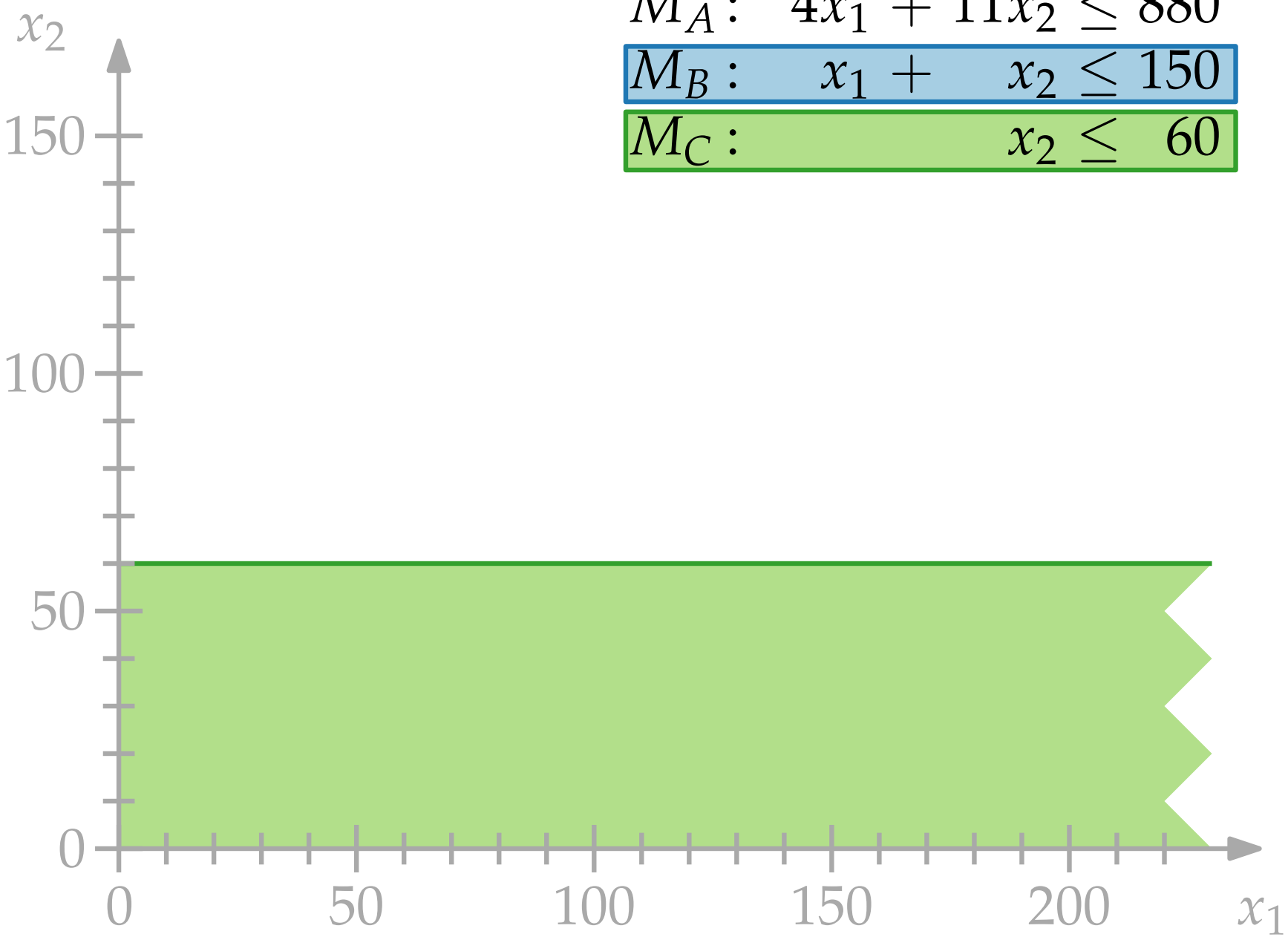
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$

$$M_B: \quad x_1 + x_2 \leq 150$$

$$M_C: \quad x_2 \leq 60$$



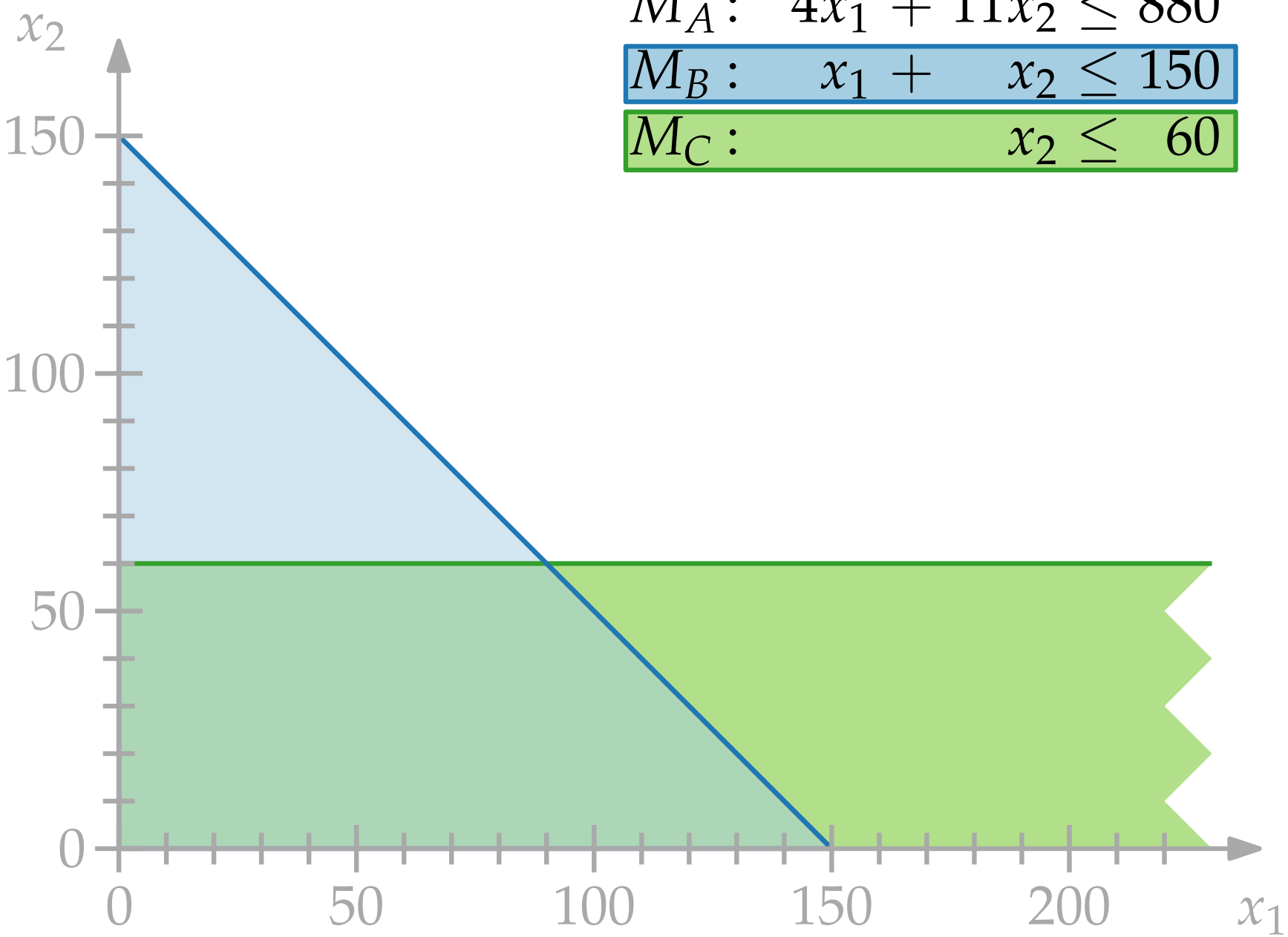
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$

$$M_B: \quad x_1 + x_2 \leq 150$$

$$M_C: \quad x_2 \leq 60$$



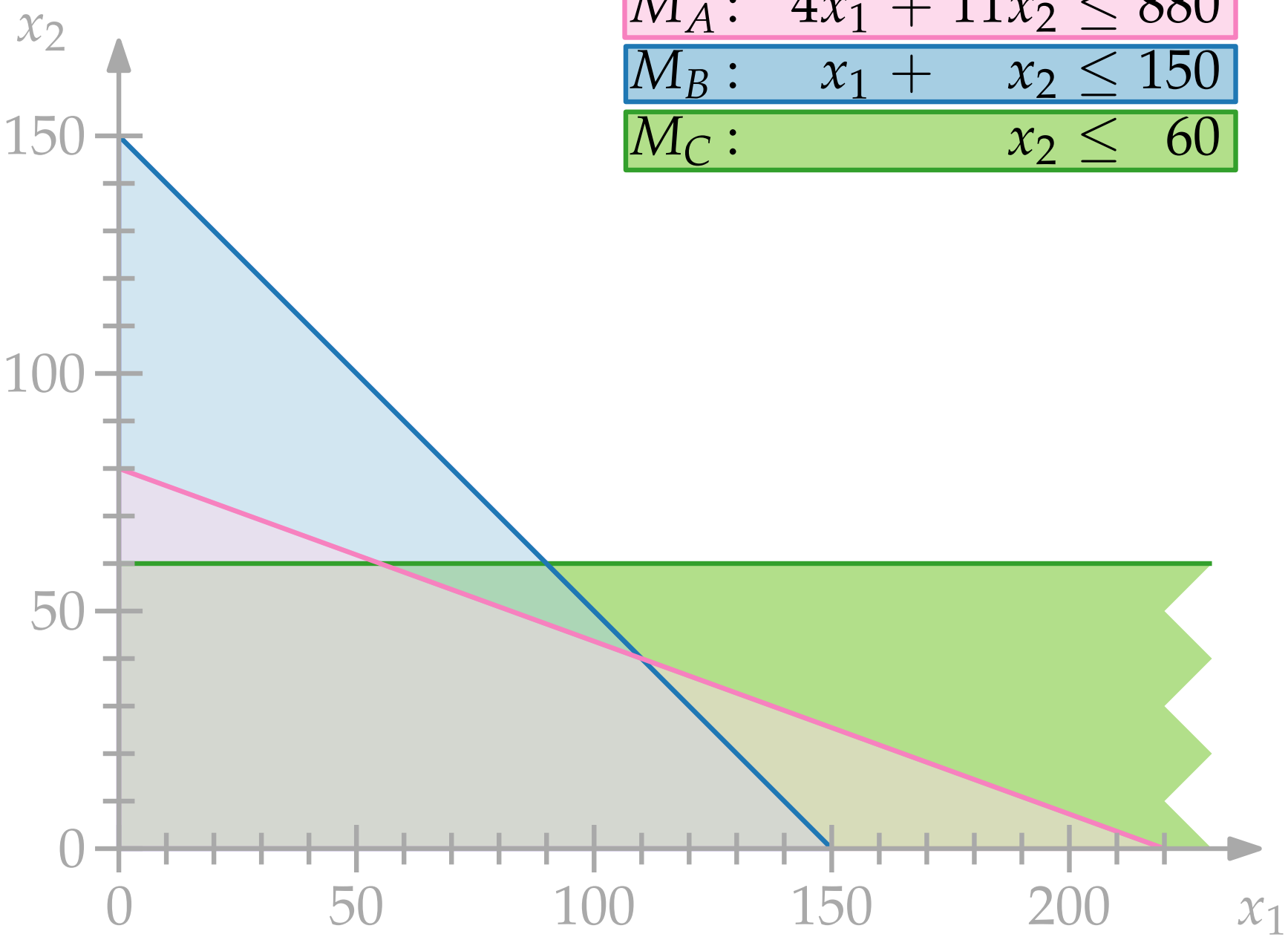
Solution

Linear constraints:

$M_A: \quad 4x_1 + 11x_2 \leq 880$

$M_B: \quad x_1 + x_2 \leq 150$

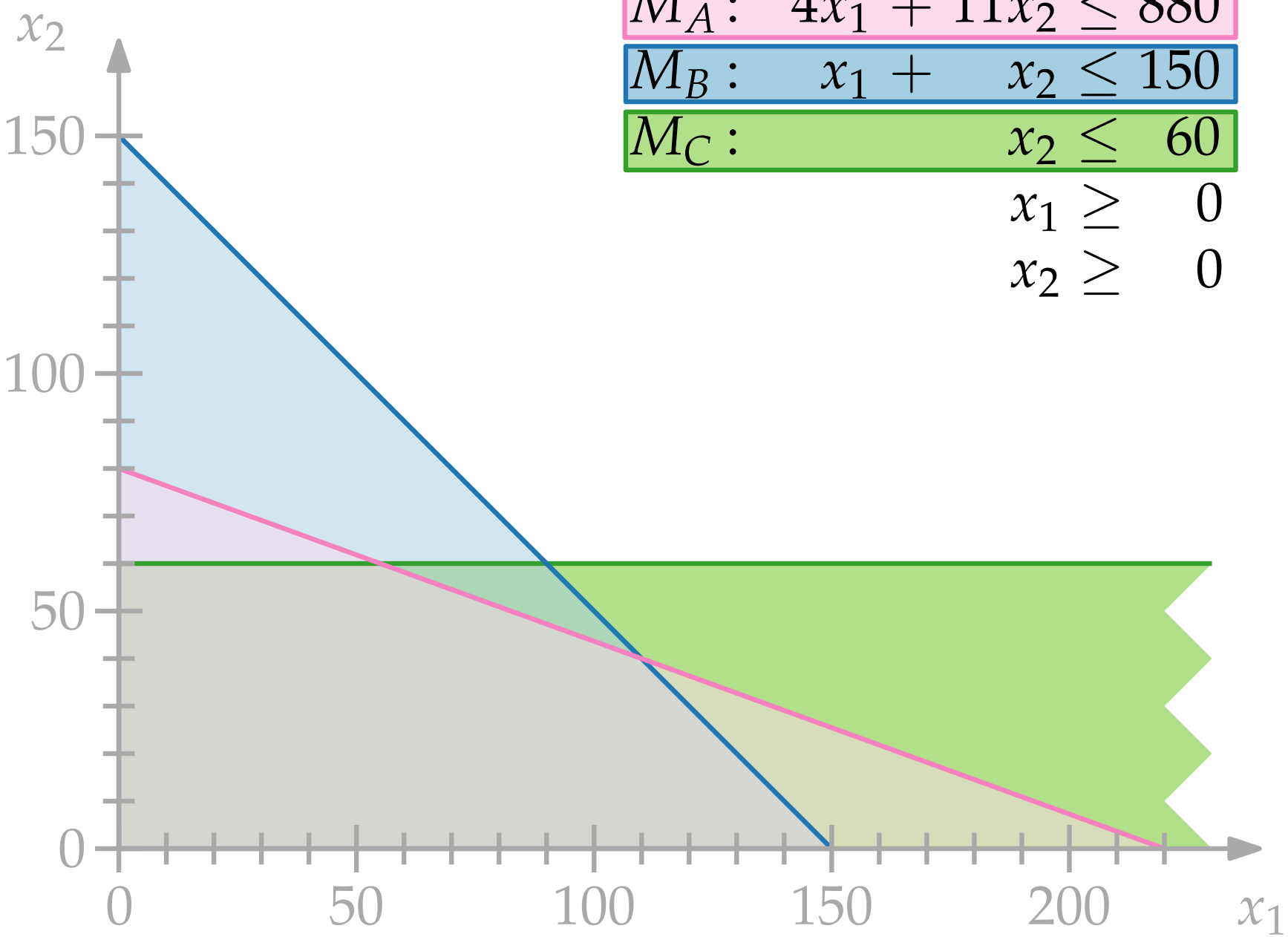
$M_C: \quad x_2 \leq 60$



Solution

Linear constraints:

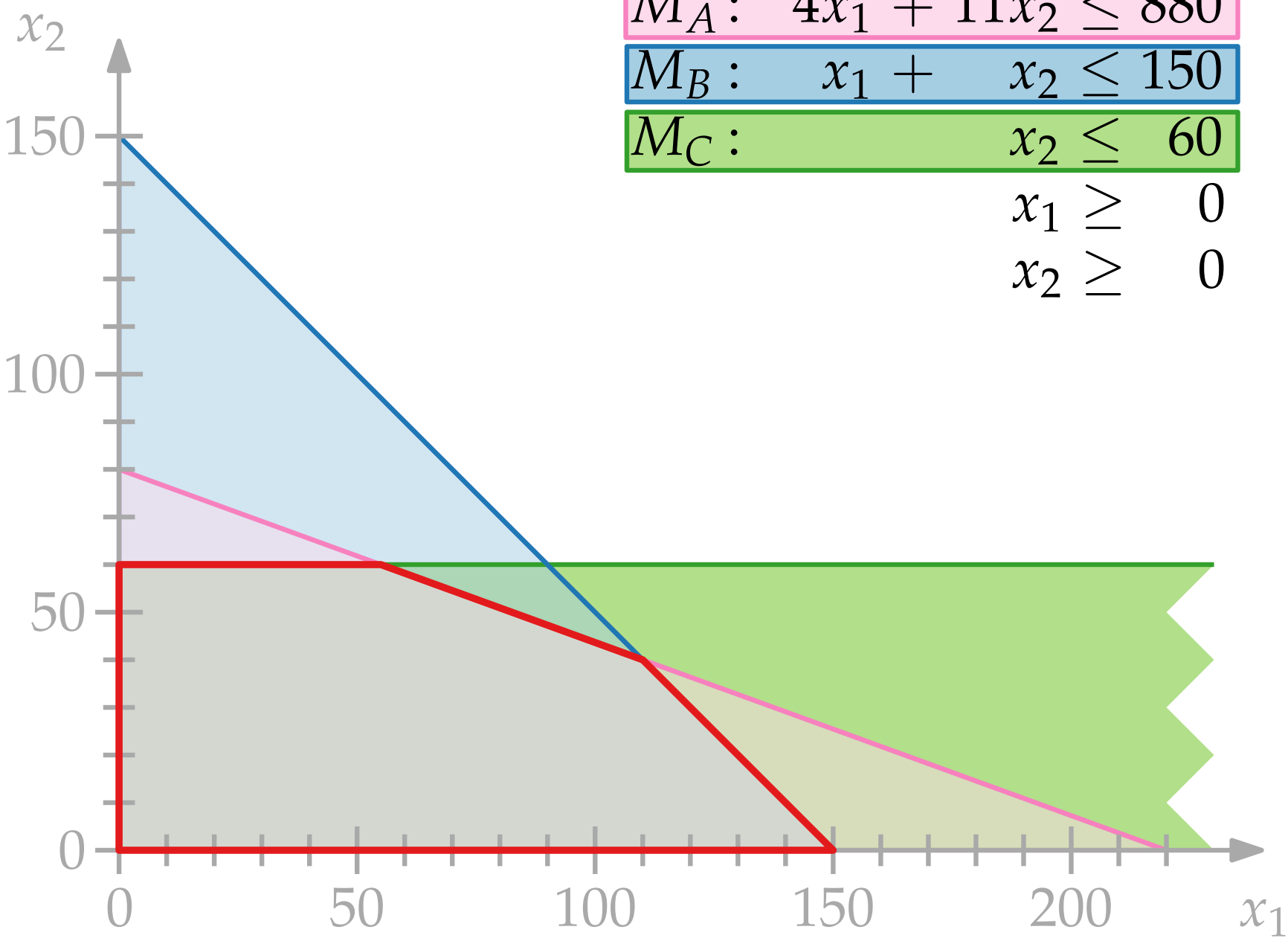
$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$



Solution

Linear constraints:

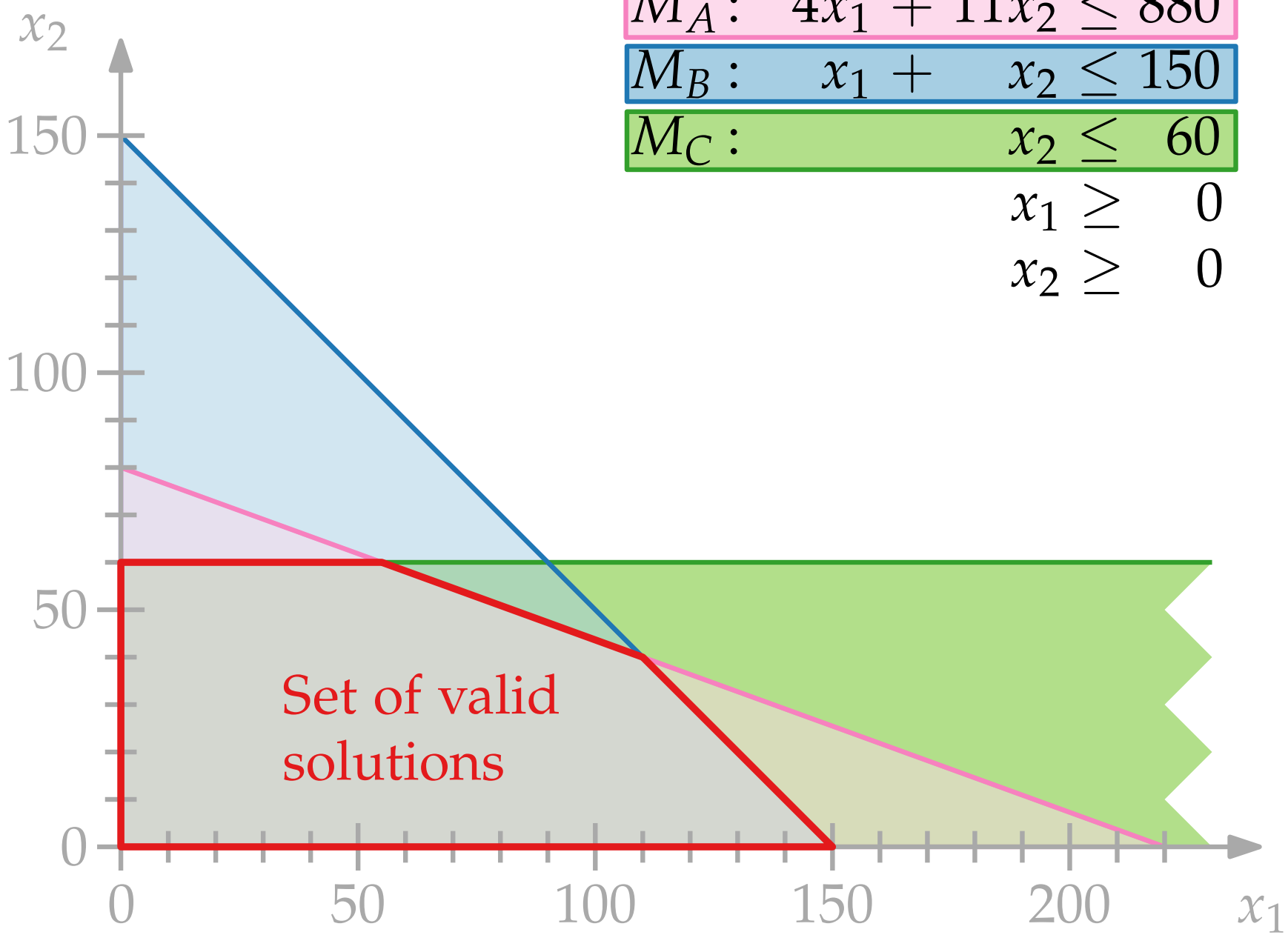
$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$



Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$



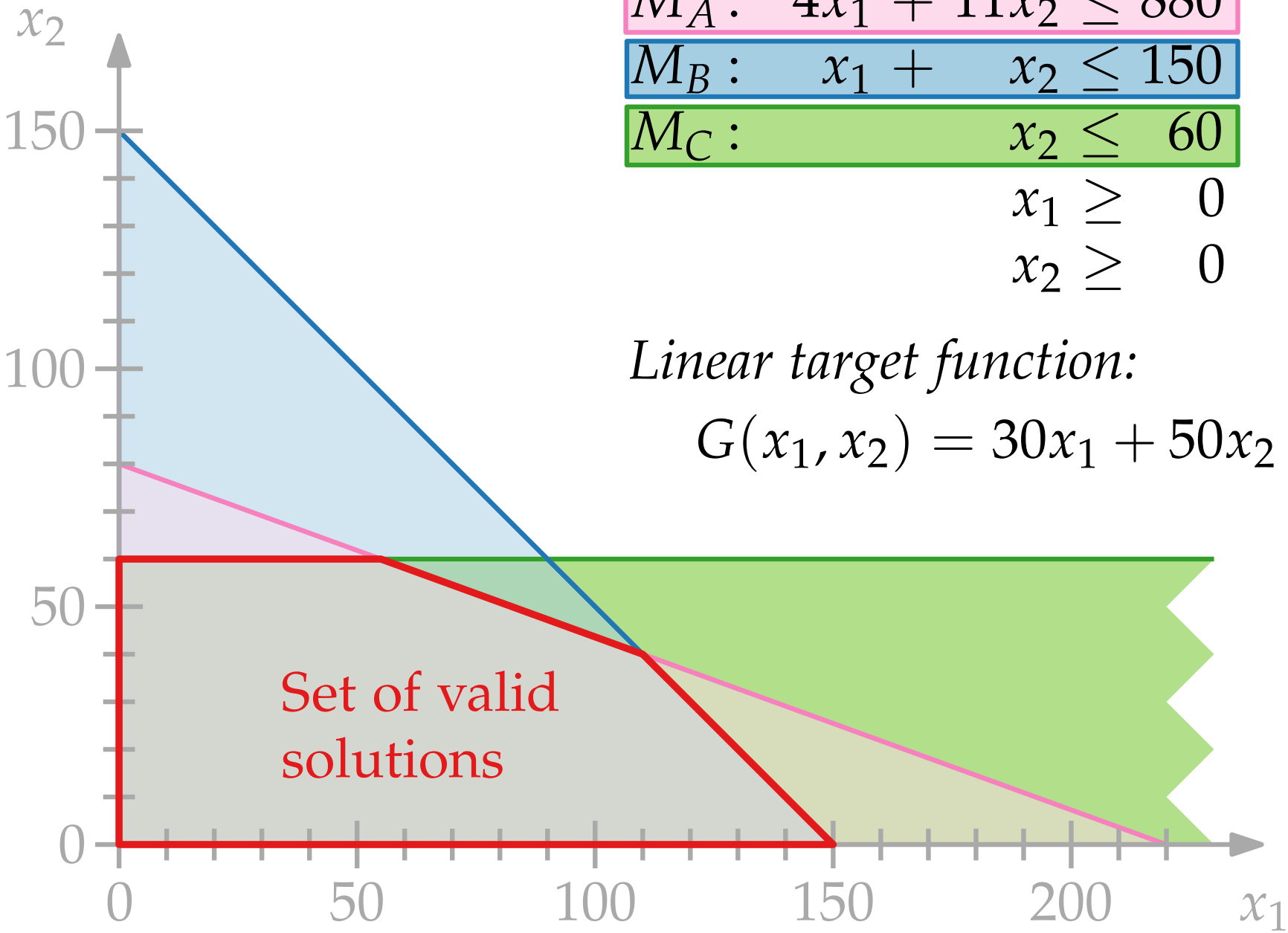
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2$$



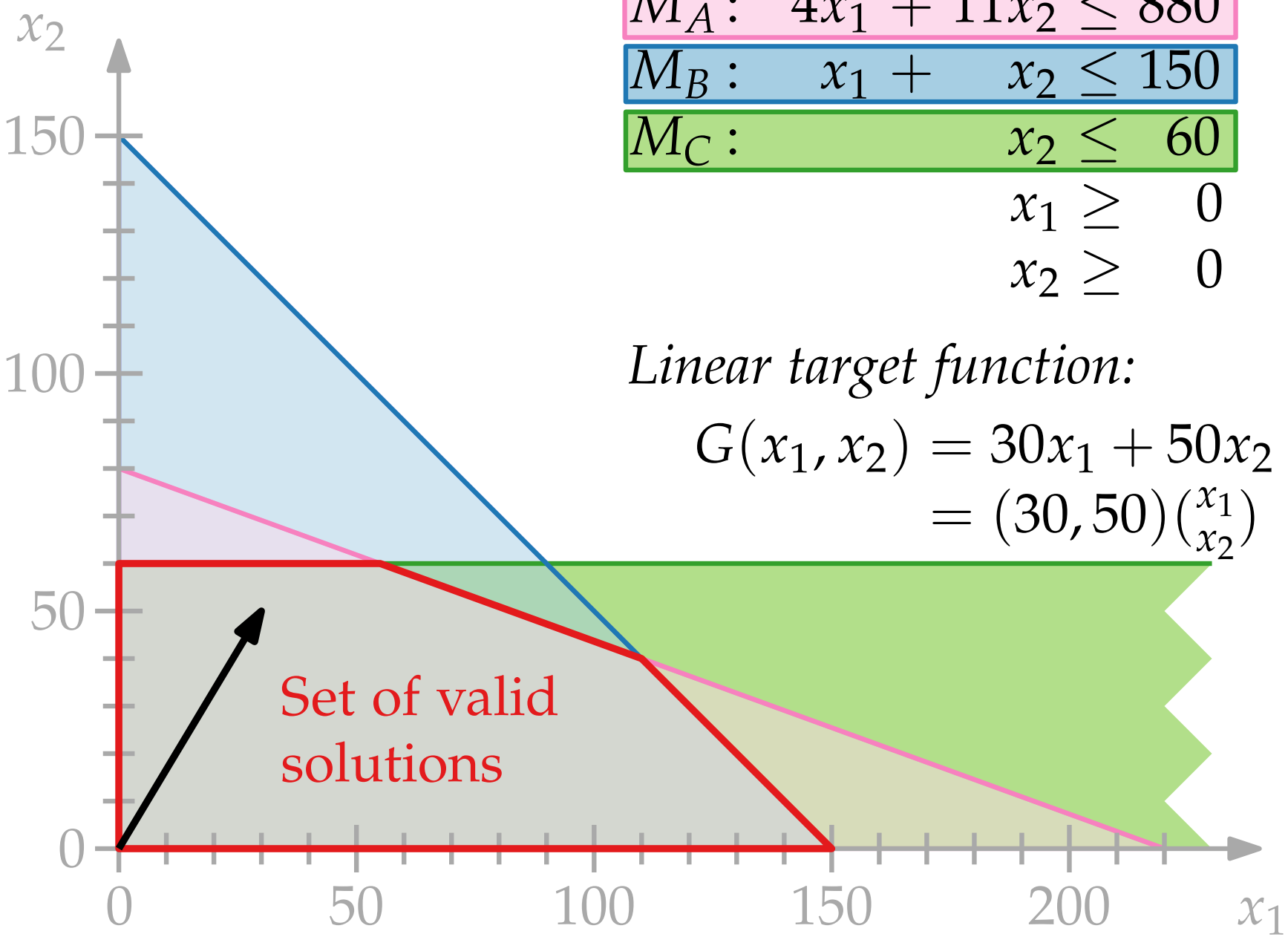
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2$$
$$= (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



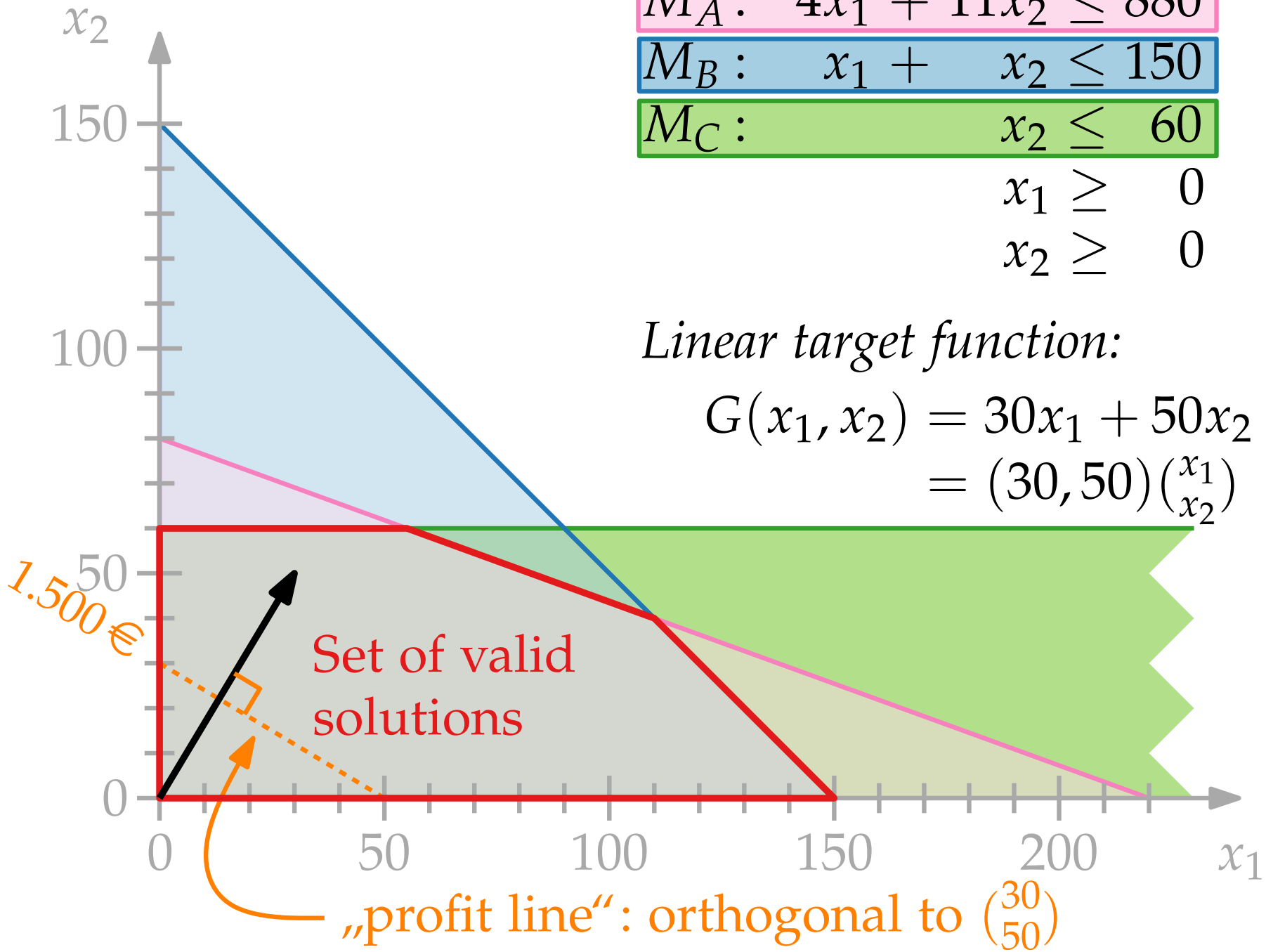
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



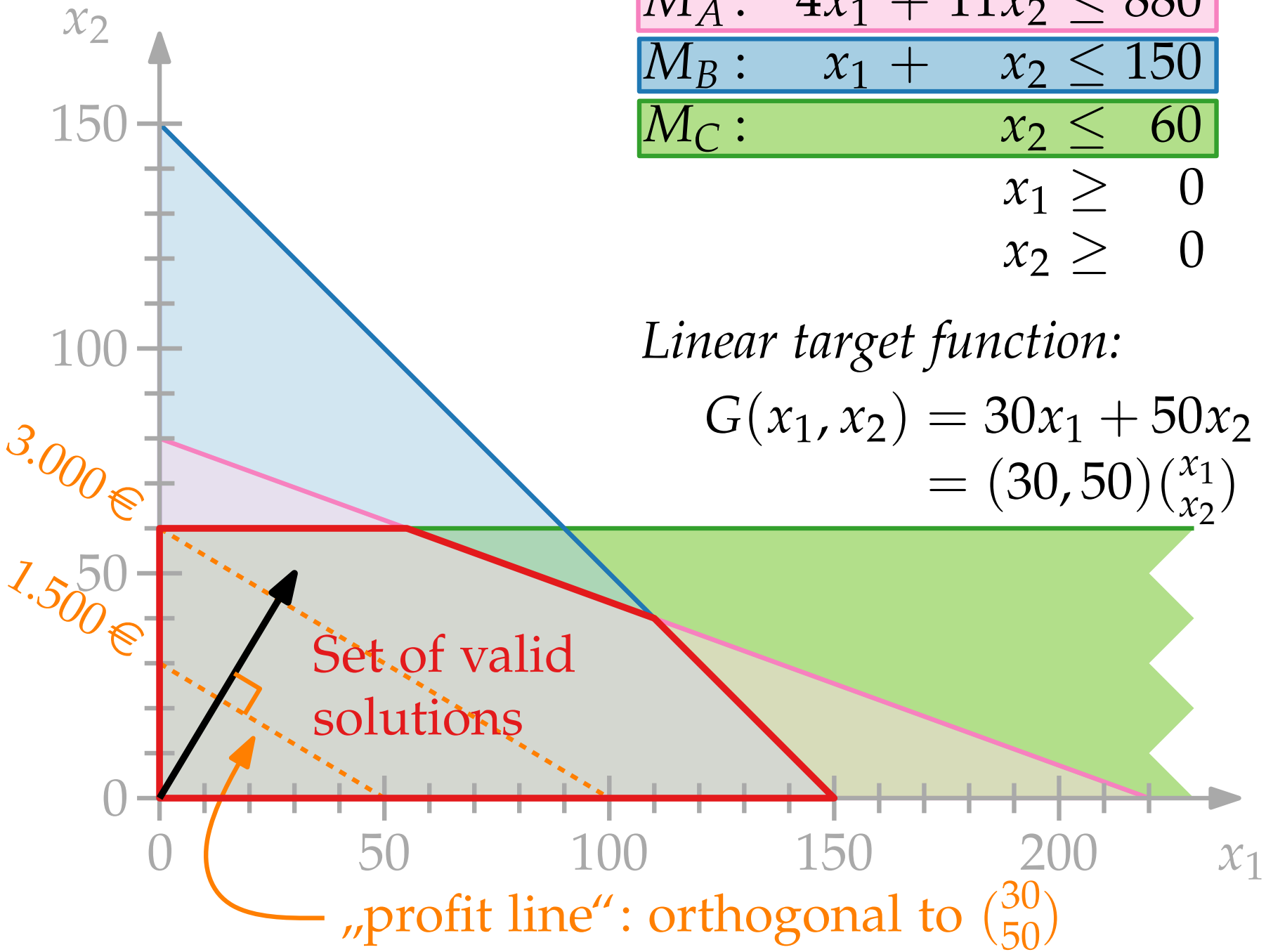
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



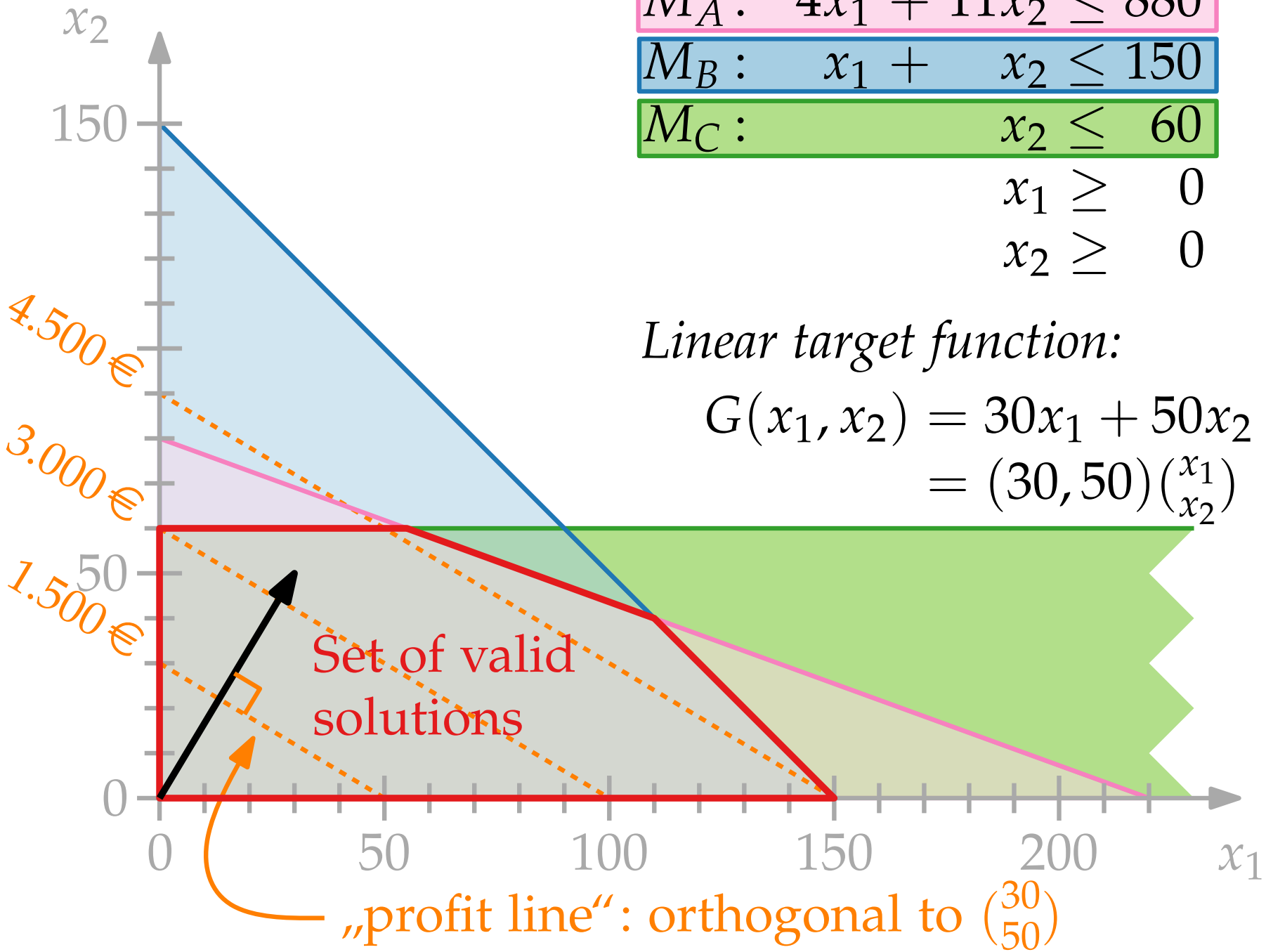
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



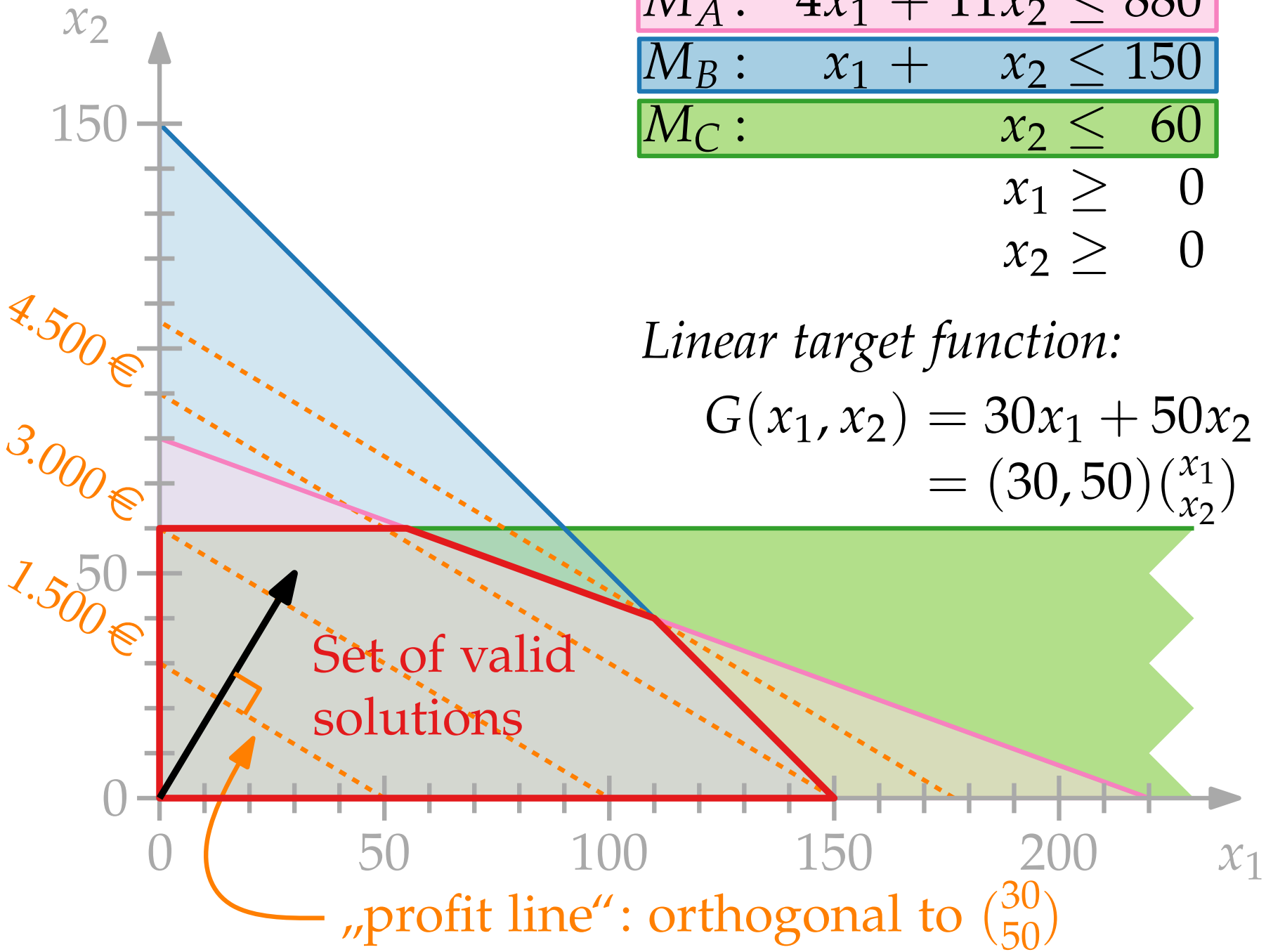
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



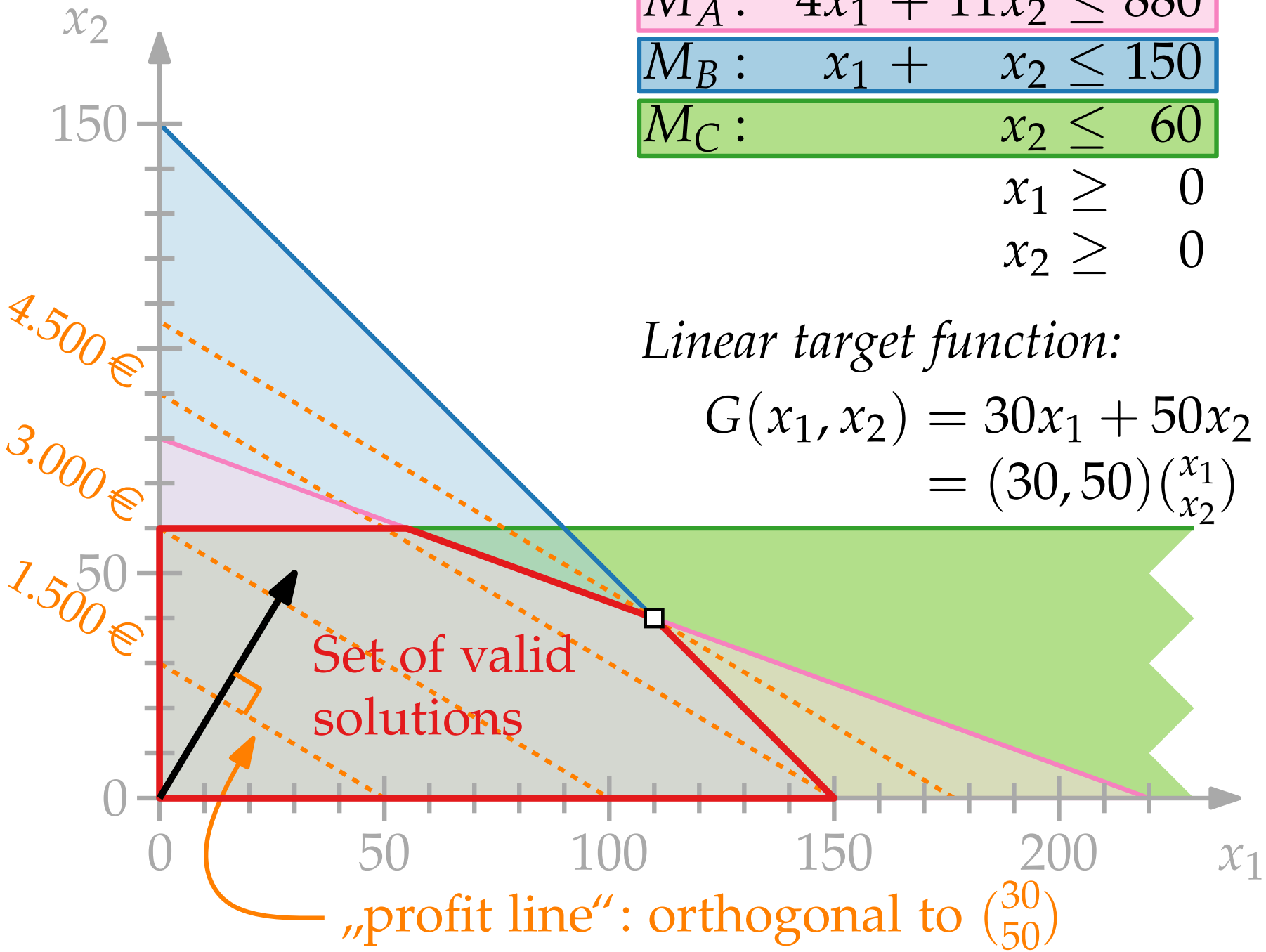
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



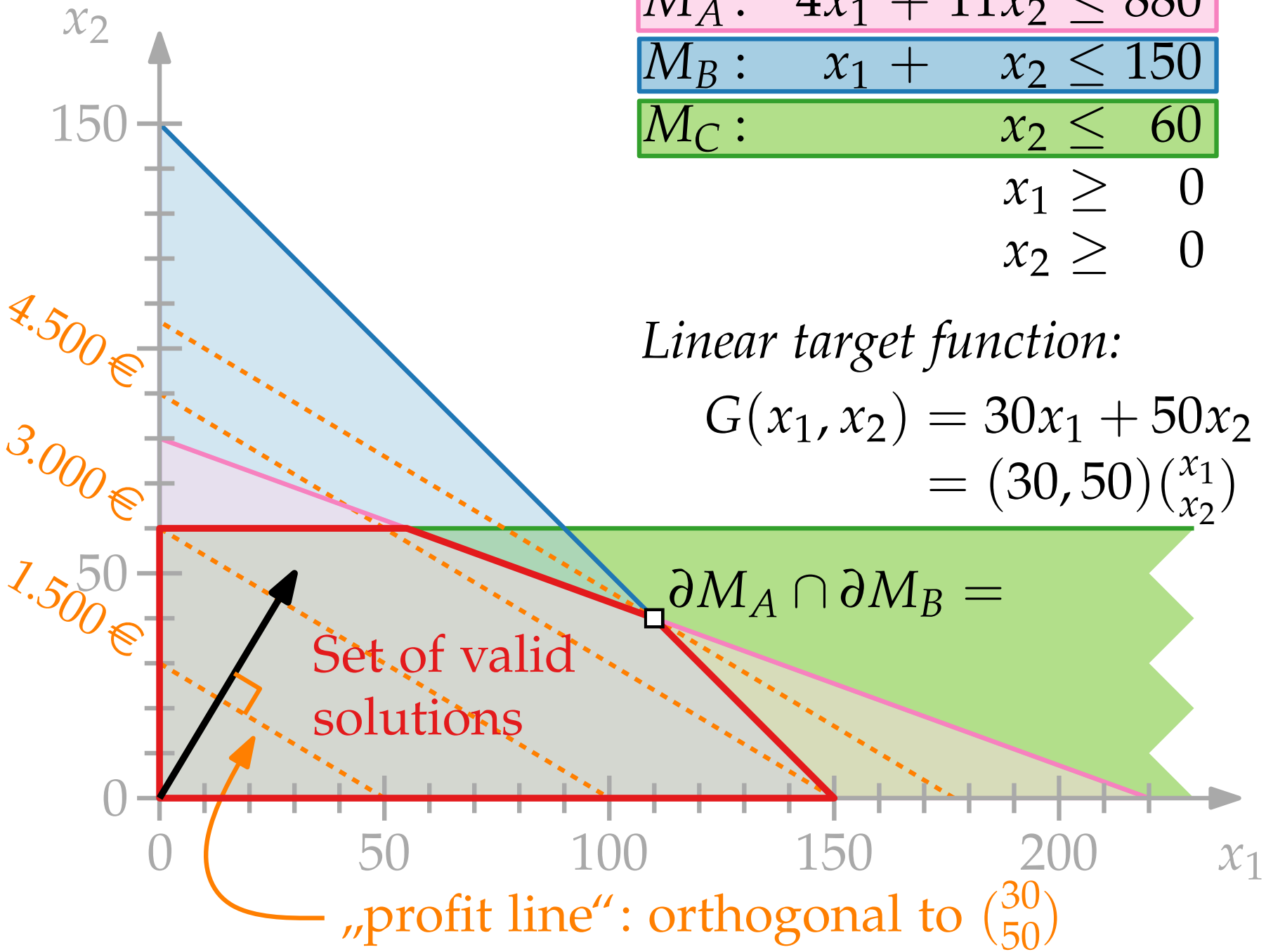
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



Solution

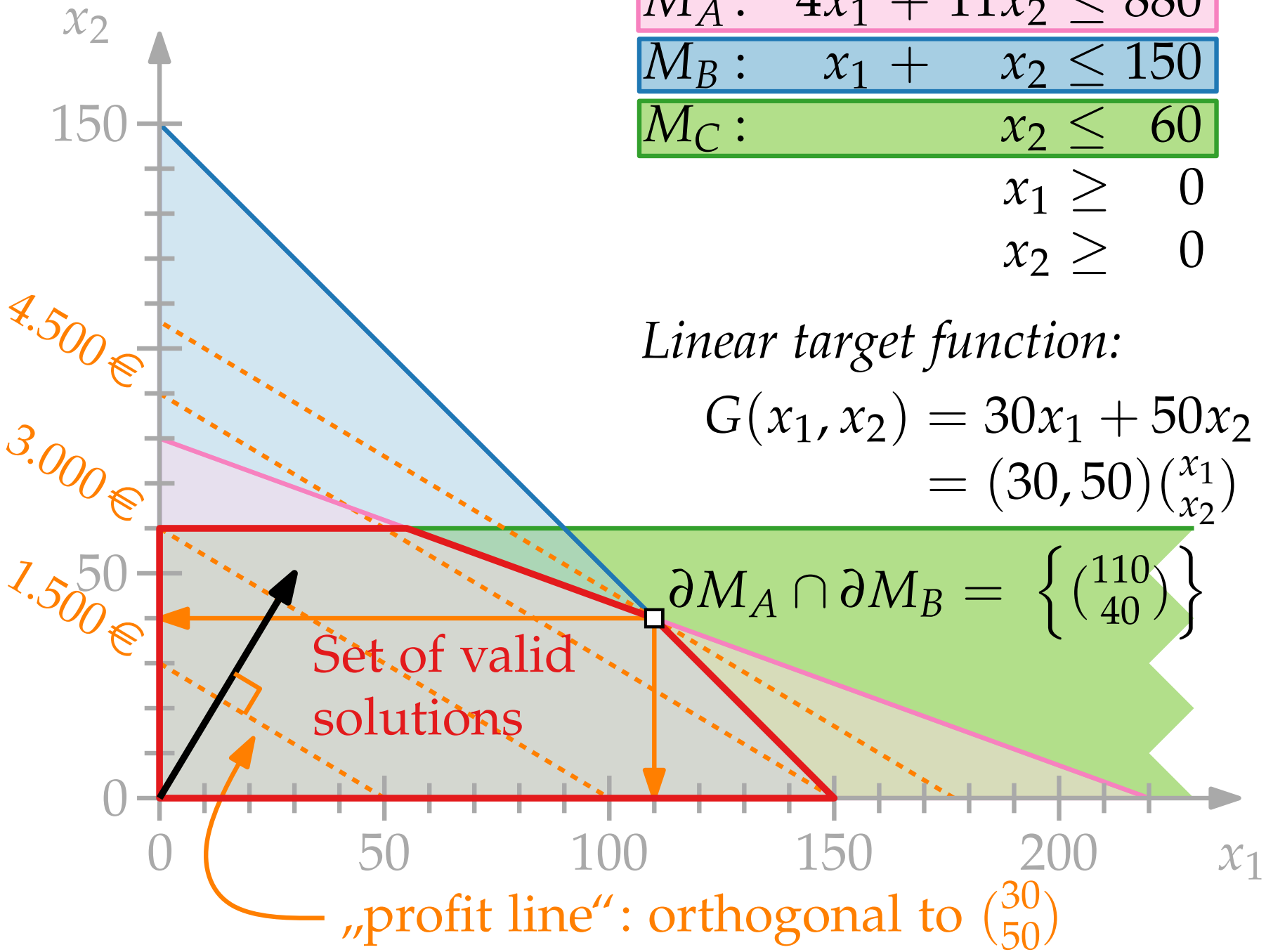
Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\partial M_A \cap \partial M_B = \left\{ \begin{pmatrix} 110 \\ 40 \end{pmatrix} \right\}$$



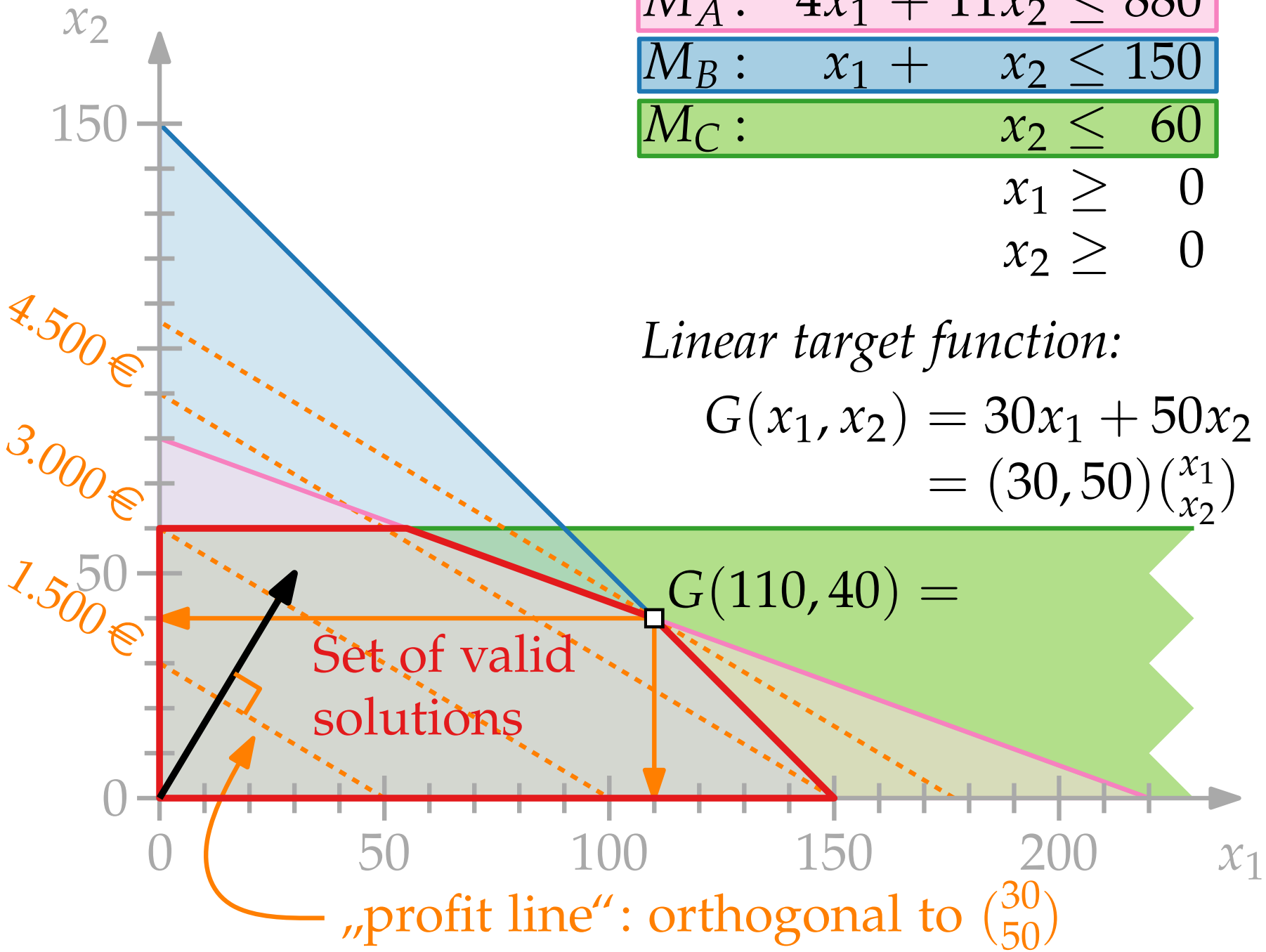
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



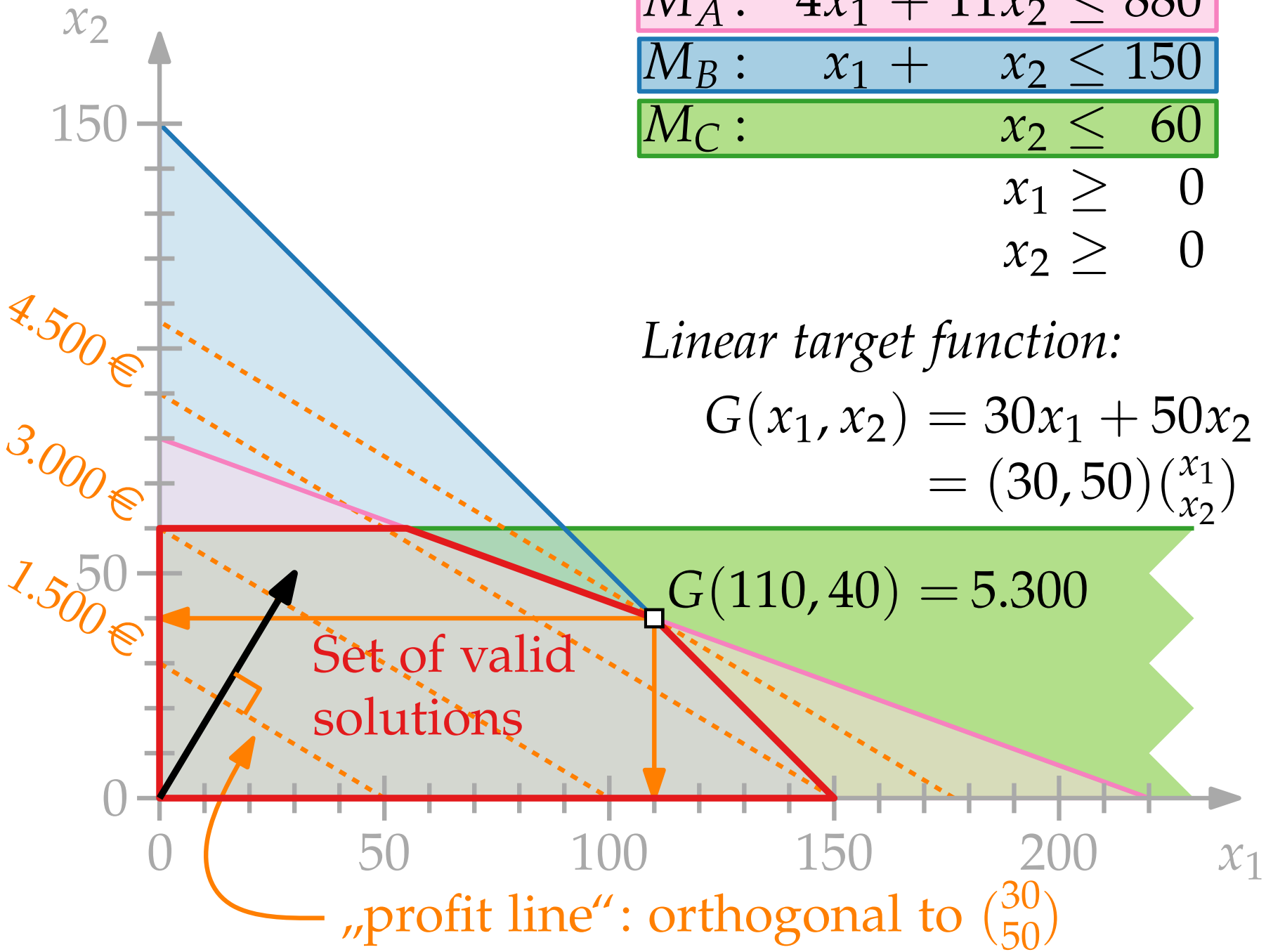
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



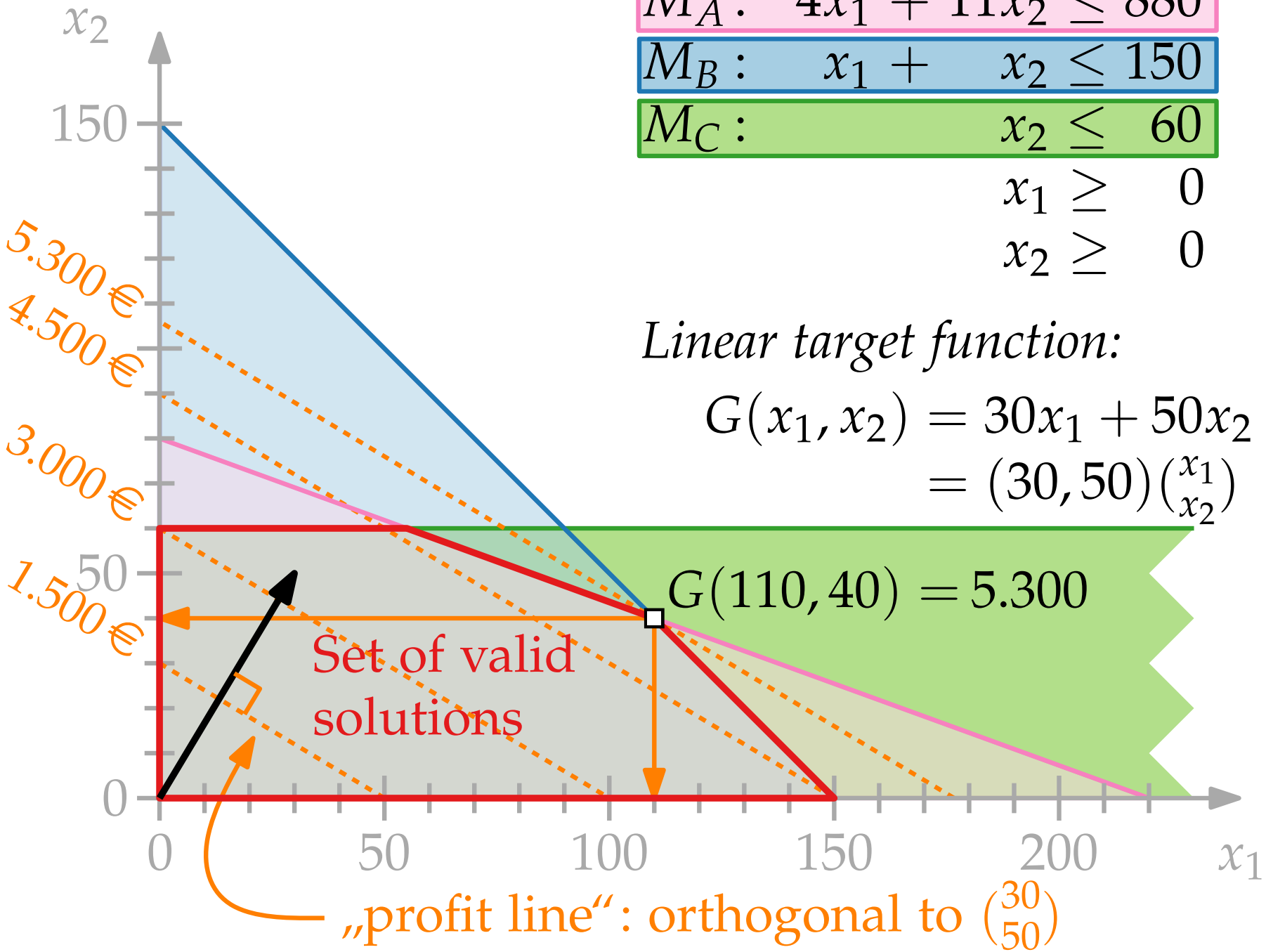
Solution

Linear constraints:

$$M_A: \quad 4x_1 + 11x_2 \leq 880$$
$$M_B: \quad x_1 + x_2 \leq 150$$
$$M_C: \quad x_2 \leq 60$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

Linear target function:

$$G(x_1, x_2) = 30x_1 + 50x_2 = (30, 50)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

Standard form (HW)

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

minimize $c^\intercal x$ Standard form (HW)

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

minimize	$c^\intercal x$			Standard form (HW)
subject to	Ax	$\geq$	b	

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

minimize	$c^\intercal x$			Standard form (HW)
subject to	Ax	$\geq$	b	
	x	$\geq$	0	

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

	minimize	$c^\intercal x$		Standard form (HW)
	subject to	Ax	$\geq$	b
		x	$\geq$	0

Example. $c = \begin{pmatrix} 7 \\ 1 \\ 5 \end{pmatrix}$ $A = \begin{pmatrix} 1 & -1 & 3 \\ 5 & 2 & -1 \end{pmatrix}$ $b = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

	minimize	$c^\intercal x$		Standard form (HW)
	subject to	Ax	$\geq$	b
		x	$\geq$	0

Example. $c = \begin{pmatrix} 7 \\ 1 \\ 5 \end{pmatrix}$ $A = \begin{pmatrix} 1 & -1 & 3 \\ 5 & 2 & -1 \end{pmatrix}$ $b = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$

minimize $7x_1 + x_2 + 5x_3$

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$\begin{array}{lrcl} & \textbf{minimize} & c^\mathsf{T} x & \text{Standard form (HW)} \\ & \textbf{subject to} & Ax & \geq \quad b \\ & & x & \geq \quad 0 \end{array}$$

Example. $c = \begin{pmatrix} 7 \\ 1 \\ 5 \end{pmatrix} \quad A = \begin{pmatrix} 1 & -1 & 3 \\ 5 & 2 & -1 \end{pmatrix} \quad b = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$

$$\begin{array}{lrcrcrcl} \textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\ \textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \end{array}$$

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$\begin{array}{lrcl} & \textbf{minimize} & c^\intercal x & \text{Standard form (HW)} \\ & \textbf{subject to} & Ax & \geq \quad b \\ & & x & \geq \quad 0 \end{array}$$

Example. $c = \begin{pmatrix} 7 \\ 1 \\ 5 \end{pmatrix} \quad A = \begin{pmatrix} 1 & -1 & 3 \\ 5 & 2 & -1 \end{pmatrix} \quad b = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$

$$\begin{array}{lrcrcrcl} \textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\ \textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\ & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \end{array}$$

Linear Programming

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

	minimize	$c^\mathsf{T} x$		Standard form (HW)
	subject to	Ax	$\geq$	b
		x	$\geq$	0

Example. $c = \begin{pmatrix} 7 \\ 1 \\ 5 \end{pmatrix}$ $A = \begin{pmatrix} 1 & -1 & 3 \\ 5 & 2 & -1 \end{pmatrix}$ $b = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$

minimize	$7x_1$	$+$	x_2	$+$	$5x_3$			
subject to	x_1	$-$	x_2	$+$	$3x_3$	$\geq$	10	
	$5x_1$	$+$	$2x_2$	$-$	x_3	$\geq$	6	
					x_1, x_2, x_3	$\geq$	0	

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 2 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & x_1, x_2, x_3 & & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 2 & - & x_2 1 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & x_1, x_2, x_3 & & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 2 & - & x_2 1 & + & 3x_3 9 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & x_1, x_2, x_3 & & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 2 & - & x_2 1 & + & 3x_3\,9 & \geq & 1010 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 2 & - & x_2 1 & + & 3x_3 9 & \geq & 10 10 \\
& 5x_1 10 & + & 2x_2 2 & - & x_3 3 & \geq & 6 9 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 14 + & & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 2 & - & x_2 1 & + & 3x_3 9 & \geq & 10 10 \\
& 5x_1 10 + & & 2x_2 2 & - & x_3 3 & \geq & 6 9 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & 14+ & x_2 & 1 & + & 5x_3 & \\
\textbf{subject to} & x_1 & 2 & - & x_2 & 1 + & 3x_3 & 9 \geq 10 10 \\
 & 5x_1 & 10+ & 2x_2 & 2 & - & x_3 & 3 \geq 6 9 \\
 & & & & & x_1, x_2, x_3 & & \geq 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & 14 + & x_2 & 1 + & 5x_3 & 15 & \\
\textbf{subject to} & x_1 & 2 - & x_2 & 1 + & 3x_3 & 9 \geq & 10\,10 \\
 & 5x_1 & 10 + & 2x_2 & 2 - & x_3 & 3 \geq & 6\,9 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & 14 + & x_2 & 1 + & 5x_3 & 15 = & 30 \\
\textbf{subject to} & x_1 & 2 - & x_2 & 1 + & 3x_3 & 9 \geq & 10\,10 \\
 & 5x_1 & 10 + & 2x_2 & 2 - & x_3 & 3 \geq & 6\,9 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

Linear Programming – Upper Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & 14+ & x_2 & 1 & + & 5x_3\ 15 & = & 30 \\
\textbf{subject to} & x_1 & 2 & - & x_2 & 1 & + & 3x_3\ 9 \geq & 10\,10 \\
 & 5x_1 & 10+ & 2x_2 & 2 & - & x_3\ 3 \geq & & 6\,9 \\
 & & & & & x_1, x_2, x_3 & \geq & & 0
\end{array}
$$

Valid solution?

$x = (2, 1, 3)$

$\Rightarrow \text{obj}(x) = 30$ is upper bound for OPT

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
 & 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & x_1, x_2, x_3 & & \geq & 0
\end{array}
$$

$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \mathrm{OPT} \geq 10$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\
&\geq 10 + 6
\end{aligned}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcr}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & x_1 & - & x_2 & + & 3x_3 & \geq & 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\
&\geq 10 + 6 \qquad \Rightarrow \text{OPT} \geq 16
\end{aligned}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & 2\cdot x_1 & - & 2\cdot x_2 & + & 2\cdot 3x_3 & \geq & 2\cdot 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\
&\geq 10 + 6 \qquad\qquad \Rightarrow \text{OPT} \geq 16
\end{aligned}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & 2\cdot x_1 & - & 2\cdot x_2 & + & 2\cdot 3x_3 & \geq & 2\cdot 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\
&\geq 10 + 6 \qquad\qquad \Rightarrow \text{OPT} \geq 16
\end{aligned}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq 2 \cdot (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\
&\geq 2\cdot 10 + 6
\end{aligned}
$$

Linear Programming – Lower Bounds

Optimize (i.e., minimize or maximize) a linear (*objective*) function subject to linear inequalities (*constraints*).

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & 2\cdot x_1 & - & 2\cdot x_2 & + & 2\cdot 3x_3 & \geq & 2\cdot 10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & x_1, x_2, x_3 & & \geq & 0
\end{array}
$$

$$7x_1 + x_2 + 5x_3 \geq x_1 - x_2 + 3x_3 \Rightarrow \text{OPT} \geq 10$$

$$\begin{aligned} 7x_1 + x_2 + 5x_3 &\geq (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\ &\geq 10 + 6 \qquad \Rightarrow \text{OPT} \geq 16 \end{aligned}$$

$$\begin{aligned} 7x_1 + x_2 + 5x_3 &\geq 2 \cdot (x_1 - x_2 + 3x_3) + (5x_1 + 2x_2 - x_3) \\ &\geq 2\cdot 10 + 6 \qquad \Rightarrow \text{OPT} \geq 26 \end{aligned}$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & {\scriptstyle 2\cdot}x_1 & - & {\scriptstyle 2\cdot}x_2 & + & {\scriptstyle 2\cdot}3x_3 & \geq & {\scriptstyle 2\cdot}10 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq 2 \cdot (x_1 - x_2 + 3x_3) + 1 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq 2 \cdot 10 + 1 \cdot 6 \quad \Rightarrow \mathrm{OPT} \geq 2 \cdot 10 + 1 \cdot 6
\end{aligned}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& 5x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq 2 \cdot (x_1 - x_2 + 3x_3) + 1 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq 2 \cdot 10 + 1 \cdot 6 \quad \Rightarrow \mathrm{OPT} \geq 2 \cdot 10 + 1 \cdot 6
\end{aligned}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq 2 \cdot (x_1 - x_2 + 3x_3) + 1 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq 2 \cdot 10 + 1 \cdot 6 \quad \Rightarrow \mathrm{OPT} \geq 2 \cdot 10 + 1 \cdot 6
\end{aligned}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

$10y_1 + 6y_2$ is lower bound for OPT

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Bounds for y_1, y_2:

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Bounds for y_1, y_2:

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

$$
y_1 + 5y_2 \leq 7
$$

Bounds for y_1, y_2:

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Bounds for y_1, y_2:

$$
\begin{array}{rcrcl}
y_1 & + & 5y_2 & \leq & 7 \\
-y_1 & + & 2y_2 & \leq & 1
\end{array}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Bounds for y_1, y_2:

$$
\begin{array}{rcrcl}
y_1 & + & 5y_2 & \leq & 7 \\
-y_1 & + & 2y_2 & \leq & 1 \\
3y_1 & - & y_2 & \leq & 5
\end{array}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Bounds for y_1, y_2:

$$
\begin{array}{rcrcl}
y_1 & + & 5y_2 & \leq & 7 \\
-y_1 & + & 2y_2 & \leq & 1 \\
3y_1 & - & y_2 & \leq & 5 \\
& & y_1, y_2 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

$$
\begin{array}{lrcrcl}
\textbf{maximize} & 10y_1 & + & 6y_2 & & \\
\textbf{subject to} & y_1 & + & 5y_2 & \leq & 7 \\
& -y_1 & + & 2y_2 & \leq & 1 \\
& 3y_1 & - & y_2 & \leq & 5 \\
& & & y_1, y_2 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Primal

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

$$
\begin{array}{lrcrcl}
\textbf{maximize} & 10y_1 & + & 6y_2 & & \\
\textbf{subject to} & y_1 & + & 5y_2 & \leq & 7 \\
& -y_1 & + & 2y_2 & \leq & 1 \\
& 3y_1 & - & y_2 & \leq & 5 \\
& & & y_1, y_2 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Primal

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Dual

$$
\begin{array}{lrcrcl}
\textbf{maximize} & 10y_1 & + & 6y_2 & & \\
\textbf{subject to} & y_1 & + & 5y_2 & \leq & 7 \\
& -y_1 & + & 2y_2 & \leq & 1 \\
& 3y_1 & - & y_2 & \leq & 5 \\
& & & y_1, y_2 & \geq & 0
\end{array}
$$

Linear Programming – Lower Bounds

Primal

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Dual

$$
\begin{array}{lrcrcl}
\textbf{maximize} & 10y_1 & + & 6y_2 & & \\
\textbf{subject to} & y_1 & + & 5y_2 & \leq & 7 \\
& -y_1 & + & 2y_2 & \leq & 1 \\
& 3y_1 & - & y_2 & \leq & 5 \\
& & & y_1, y_2 & \geq & 0
\end{array}
$$

Any feasible solution to the **dual** program provides a lower bound for the optimum of the **primal** program.

Linear Programming – Lower Bounds

Primal

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Dual

$$
\begin{array}{lrcrcl}
\textbf{maximize} & 10y_1 & + & 6y_2 & & \\
\textbf{subject to} & y_1 & + & 5y_2 & \leq & 7 \\
& -y_1 & + & 2y_2 & \leq & 1 \\
& 3y_1 & - & y_2 & \leq & 5 \\
& & & y_1, y_2 & \geq & 0
\end{array}
$$

Any feasible solution to the **dual** program provides a lower bound for the optimum of the **primal** program.

$x = (7/4, 0, 11/4)$ both $y = (2, 1)$ provide objective value 26.

Linear Programming – Lower Bounds

Primal

$$
\begin{array}{lrcrcrcl}
\textbf{minimize} & 7x_1 & + & x_2 & + & 5x_3 & & \\
\textbf{subject to} & y_1(x_1 & - & x_2 & + & 3x_3) & \geq & 10y_1 \\
& y_2(5x_1 & + & 2x_2 & - & x_3) & \geq & 6y_2 \\
& & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

$$
\begin{aligned}
7x_1 + x_2 + 5x_3 &\geq y_1 \cdot (x_1 - x_2 + 3x_3) + y_2 \cdot (5x_1 + 2x_2 - x_3) \\
&\geq y_1 \cdot 10 + y_2 \cdot 6 \Rightarrow \mathrm{OPT} \geq 10y_1 + 6y_2
\end{aligned}
$$

Dual

$$
\begin{array}{lrcrcl}
\textbf{maximize} & 10y_1 & + & 6y_2 & & \\
\textbf{subject to} & y_1 & + & 5y_2 & \leq & 7 \\
& -y_1 & + & 2y_2 & \leq & 1 \\
& 3y_1 & - & y_2 & \leq & 5 \\
& & & y_1, y_2 & \geq & 0
\end{array}
$$

Any feasible solution to the **dual** program provides a lower bound for the optimum of the **primal** program.

$x = (7/4, 0, 11/4)$ both $y = (2, 1)$ provide objective value $26 = \mathrm{OPT}$.

Primal – Dual

Primal Program

$$
\begin{array}{lrcl}
\textbf{minimize} & c^\mathsf{T} x & & \\
\textbf{subject to} & Ax & \geq & b \\
& x & \geq & 0
\end{array}
$$

Primal – Dual

Primal Program

$$\begin{aligned} \textbf{minimize} \quad & c^\top x \\ \textbf{subject to} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

Dual Program

$$\begin{aligned} \textbf{maximize} \quad & b^\top y \\ \textbf{subject to} \quad & A^\top y \leq c \\ & y \geq 0 \end{aligned}$$

Primal – Dual

Primal Program

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \geq & 0 \end{array}$$

Dual Program

$$\begin{array}{lrcl} \textbf{maximize} & b^\top y & & \\ \textbf{subject to} & A^\top y & \leq & c \\ & y & \geq & 0 \end{array}$$

Dual Program of the Dual Program

Primal – Dual

Primal Program

$$\begin{aligned} \textbf{minimize} \quad & c^\top x \\ \textbf{subject to} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

Dual Program

$$\begin{aligned} \textbf{maximize} \quad & b^\top y \\ \textbf{subject to} \quad & A^\top y \leq c \\ & y \geq 0 \end{aligned}$$

Dual Program of the Dual Program

$$\begin{aligned} \textbf{minimize} \quad & c^\top x \\ \textbf{subject to} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

LP-Duality

Primal

$$\begin{aligned} \textbf{minimize} \quad & c^\top x & & \\ \textbf{subject to} \quad & Ax & \geq & b \\ & x & \geq & 0 \end{aligned}$$

Dual

$$\begin{aligned} \textbf{maximize} \quad & b^\top y & & \\ \textbf{subject to} \quad & A^\top y & \leq & c \\ & y & \geq & 0 \end{aligned}$$

LP-Duality

Primal

$$\begin{aligned} \textbf{minimize} \quad & c^\top x & & \\ \textbf{subject to} \quad & Ax & \geq & b \\ & x & \geq & 0 \end{aligned}$$

Dual

$$\begin{aligned} \textbf{maximize} \quad & b^\top y & & \\ \textbf{subject to} \quad & A^\top y & \leq & c \\ & y & \geq & 0 \end{aligned}$$

Theorem. The primal program has a finite optimum $\Leftrightarrow$ the dual program has a finite optimum.

LP-Duality

Primal

$$\begin{aligned} \textbf{minimize} \quad & c^\top x & & \\ \textbf{subject to} \quad & Ax & \geq & b \\ & x & \geq & 0 \end{aligned}$$

Dual

$$\begin{aligned} \textbf{maximize} \quad & b^\top y & & \\ \textbf{subject to} \quad & A^\top y & \leq & c \\ & y & \geq & 0 \end{aligned}$$

Theorem. The primal program has a finite optimum $\Leftrightarrow$ the dual program has a finite optimum. Moreover, if $x^* = (x_1^*, \ldots, x_n^*)$ and $y^* = (y_1^*, \ldots, y_m^*)$ are *optimal* solutions for the primal and dual program (resp.), then

LP-Duality

Primal

$$\begin{aligned} \textbf{minimize} \quad & c^\top x & & \\ \textbf{subject to} \quad & Ax & \geq & b \\ & x & \geq & 0 \end{aligned}$$

Dual

$$\begin{aligned} \textbf{maximize} \quad & b^\top y & & \\ \textbf{subject to} \quad & A^\top y & \leq & c \\ & y & \geq & 0 \end{aligned}$$

Theorem. The primal program has a finite optimum $\Leftrightarrow$ the dual program has a finite optimum. Moreover, if $x^* = (x_1^*, \ldots, x_n^*)$ and $y^* = (y_1^*, \ldots, y_m^*)$ are *optimal* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j^* = \sum_{i=1}^{m} b_i y_i^* \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \qquad\qquad \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \qquad\qquad \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \qquad\qquad \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \qquad \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \geq \sum_{i=1}^{m} b_i y_i \,.$$

($\geq$ over c_j)

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i x_j \right) \geq \sum_{i=1}^{m} b_i y_i \,.$$



Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \overset{\geq}{\geq} \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} y_i x_j \right) \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \overset{\geq}{\geq} \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \overset{\geq}{\geq} \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$			
subject to	Ax	$\geq$	b	
	x	$\geq$	0	

maximize	$b^\top y$			
subject to	$A^\top y$	$\leq$	c	
	y	$\geq$	0	

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize $c^\top x$
subject to $Ax \geq b$
$x \geq 0$

maximize $b^\top y$
subject to $A^\top y \leq c$
$y \geq 0$

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \overset{\geq}{\geq} \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i \,.$$

Weak LP-Duality

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ are *valid* solutions for the primal and dual program (resp.), then

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{i=1}^{m} b_i y_i \,.$$

Proof.

$$\sum_{j=1}^{n} c_j x_j \overset{\geq}{\geq} \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \overset{\geq}{\geq} \sum_{i=1}^{m} b_i y_i \,.$$

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.).

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Complementary Slackness

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \geq & 0 \end{array}$$

$$\begin{array}{lrcl} \textbf{maximize} & b^\top y & & \\ \textbf{subject to} & A^\top y & \leq & c \\ & y & \geq & 0 \end{array}$$

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Proof.

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Proof. Follows from LP-Duality:

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Proof. Follows from LP-Duality:

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i \,.$$

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Proof. Follows from LP-Duality:

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i \,.$$

Complementary Slackness

minimize	$c^\top x$		
subject to	Ax	$\geq$	b
	x	$\geq$	0

maximize	$b^\top y$		
subject to	$A^\top y$	$\leq$	c
	y	$\geq$	0

Theorem. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ be valid solutions for the primal and dual program (resp.). Then x and y are optimal if and only if the following conditions are met:

Primal CS:
For each $j = 1, \ldots, n$: either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
For each $i = 1, \ldots, m$: either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

Proof. Follows from LP-Duality:

$$\sum_{j=1}^{n} c_j x_j \geq \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} y_i \right) x_j = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j \right) y_i \geq \sum_{i=1}^{m} b_i y_i .$$

LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

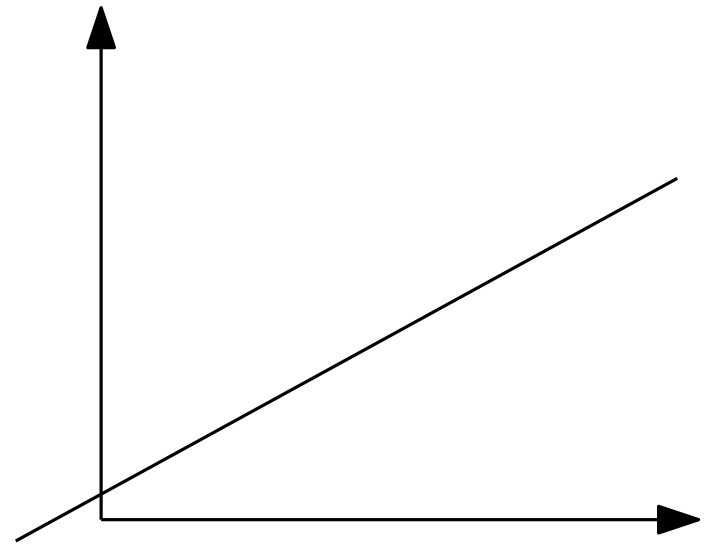
LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).



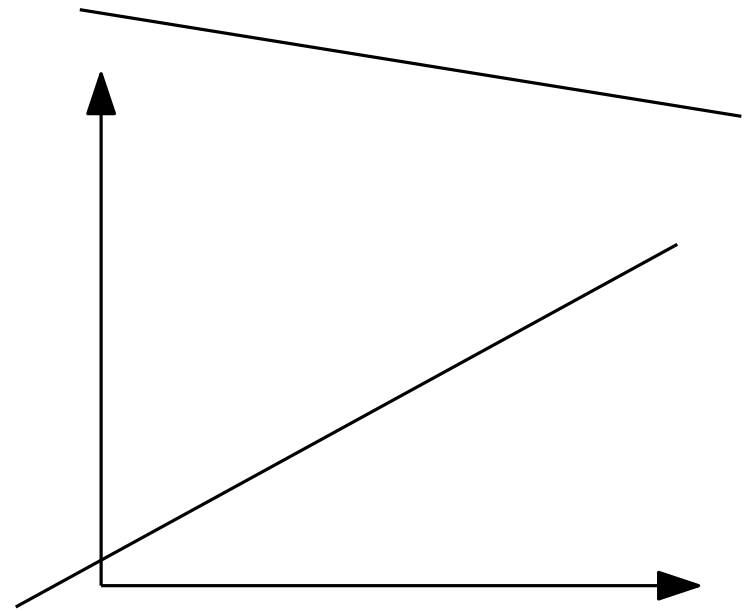
LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).



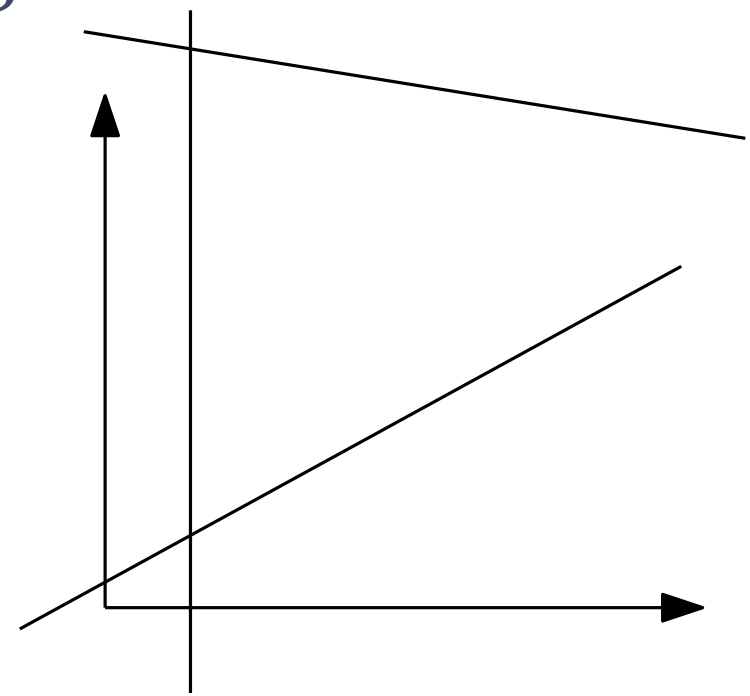
LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).



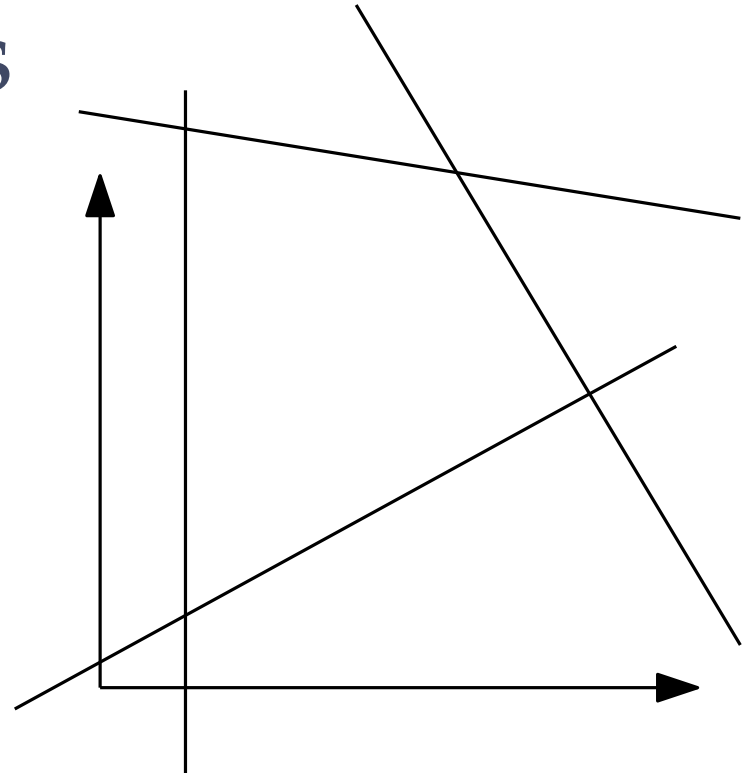
LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).



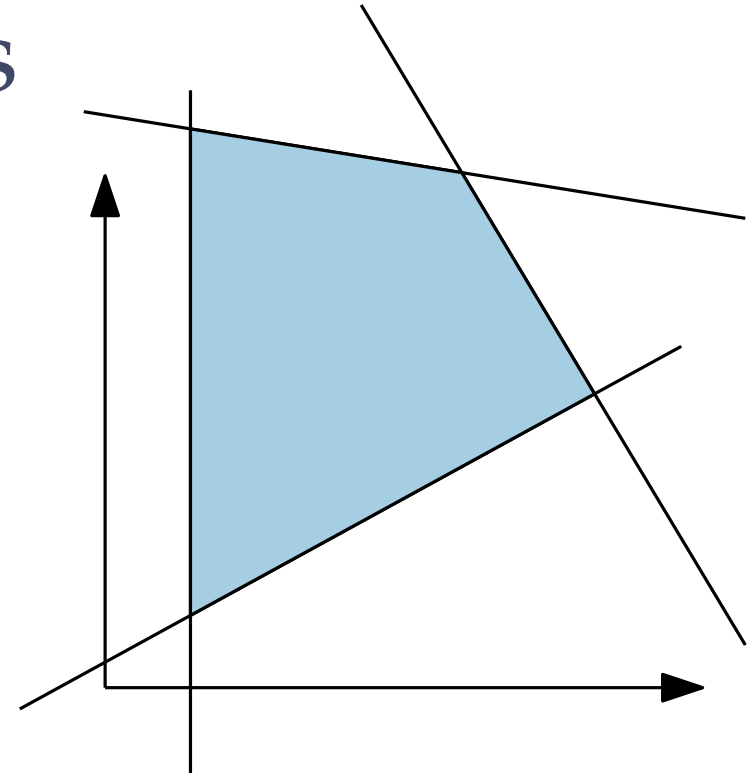
LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).



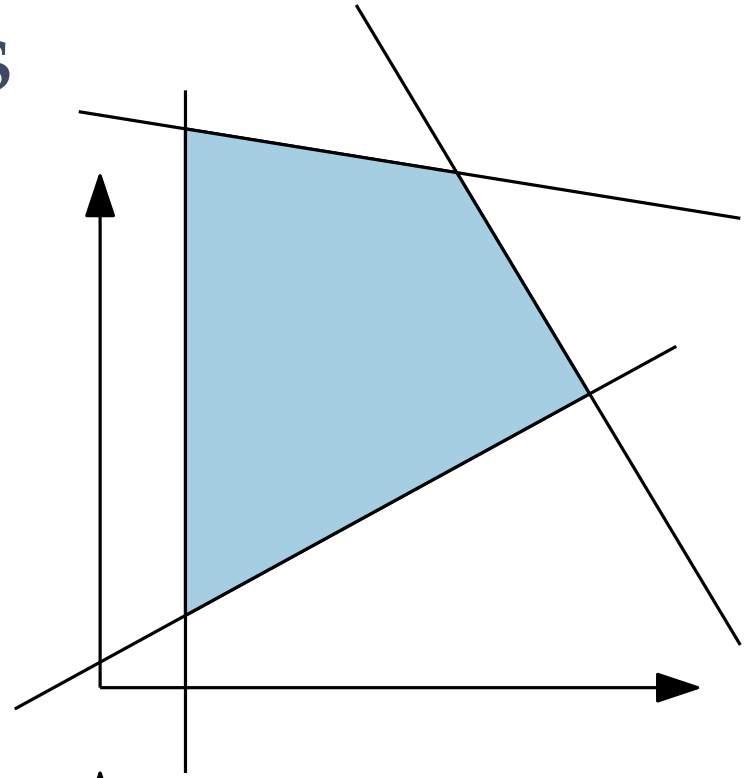
LPs and Convex Polytopes

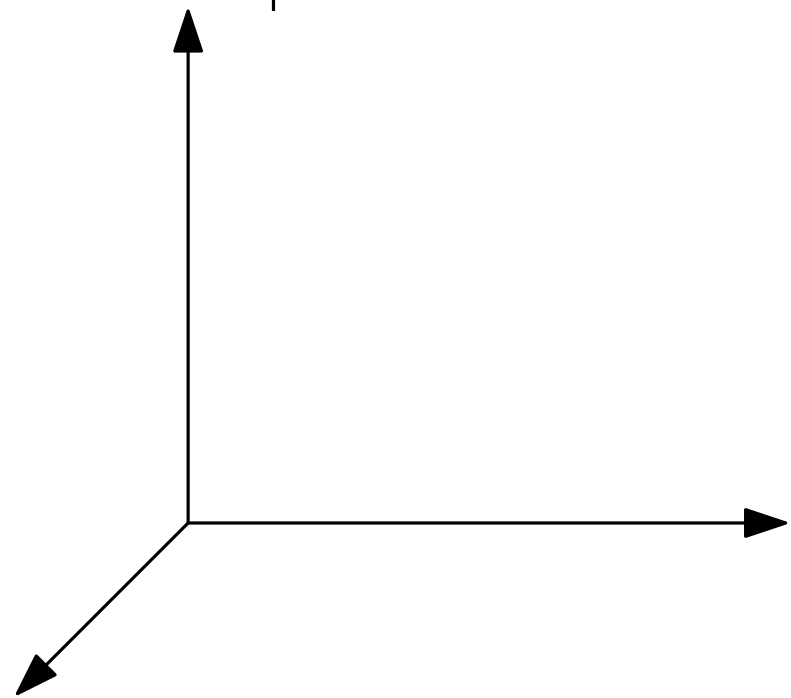
The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).



LPs and Convex Polytopes

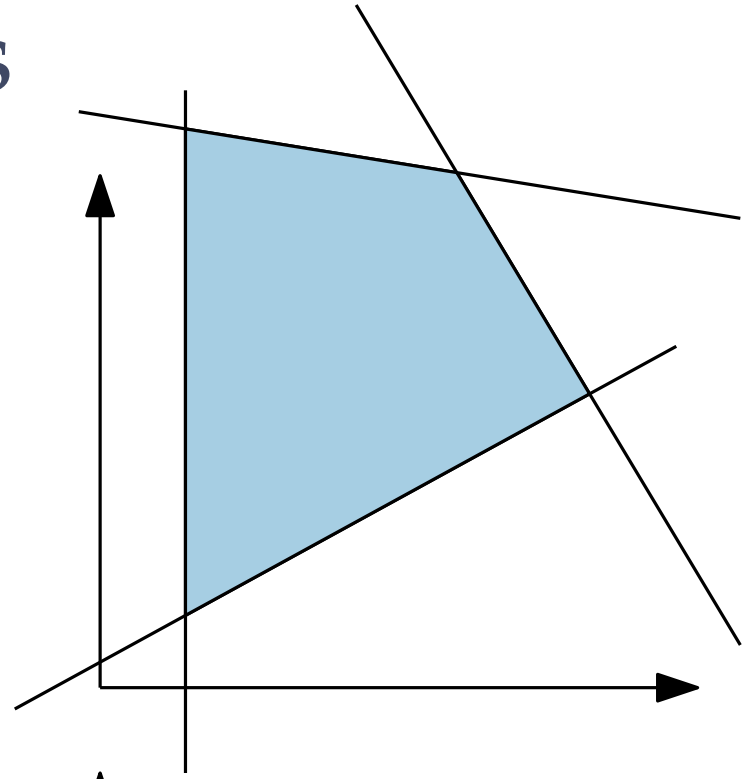
The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

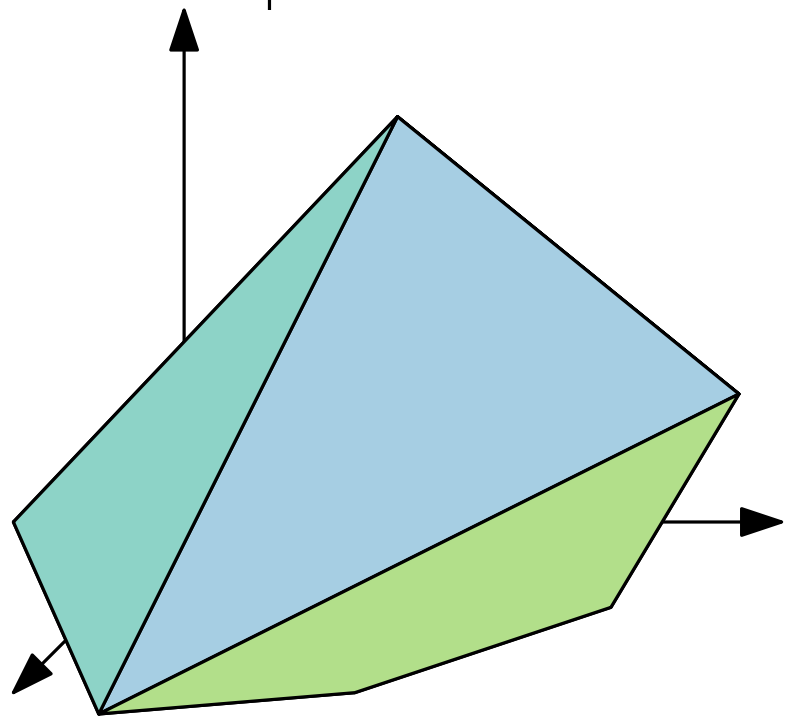




LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

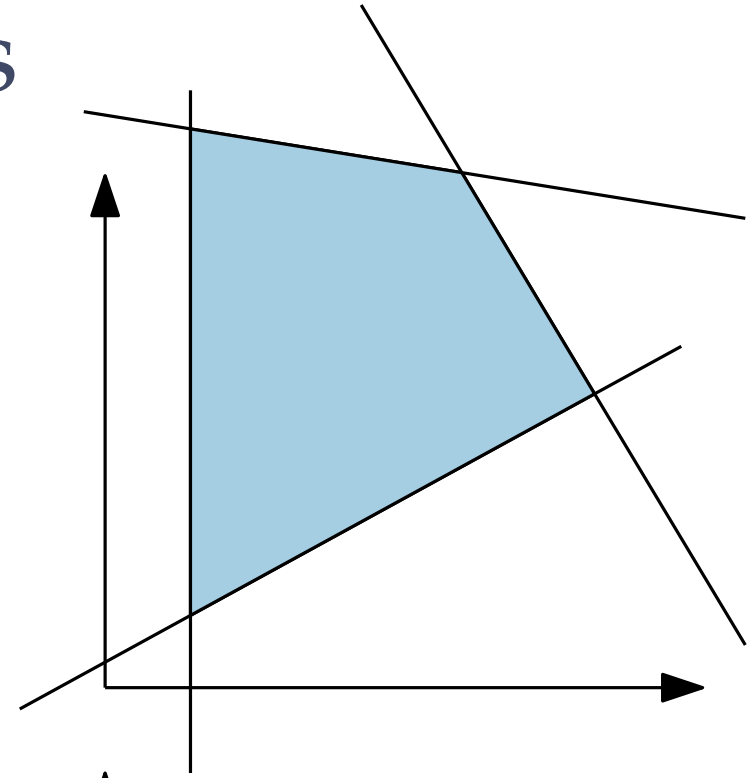


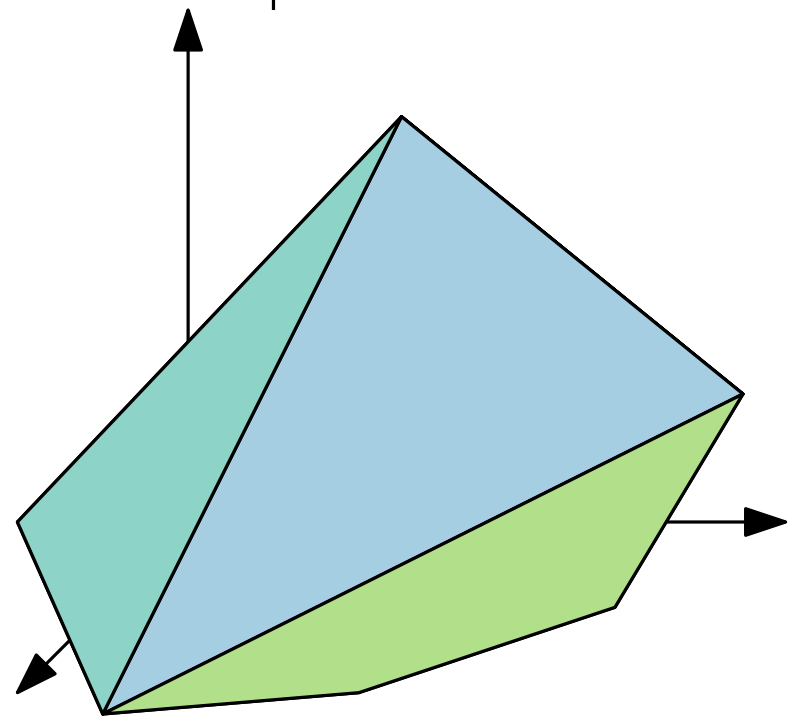


LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

Corners of the polytope are called **extreme point solutions** $\Leftrightarrow$
n linearly independent inequalities (constraints) are satisfied with equality.

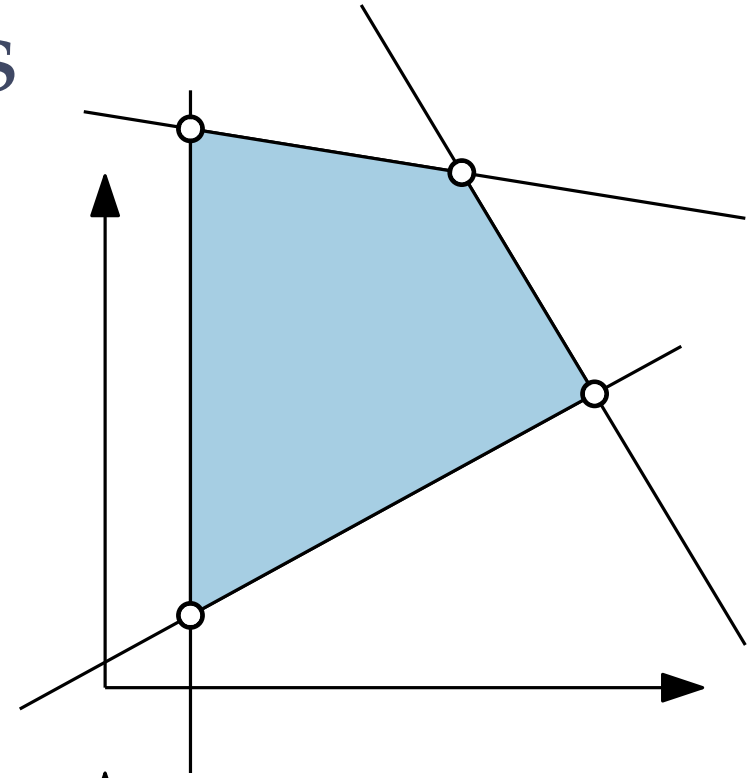


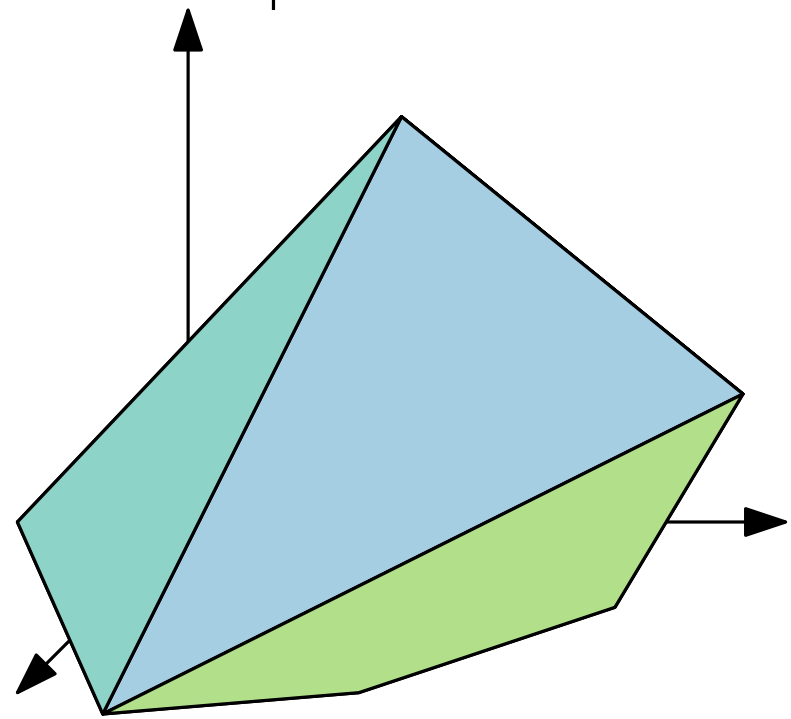


LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

Corners of the polytope are called **extreme point solutions** $\Leftrightarrow$
n linearly independent inequalities (constraints) are satisfied with equality.

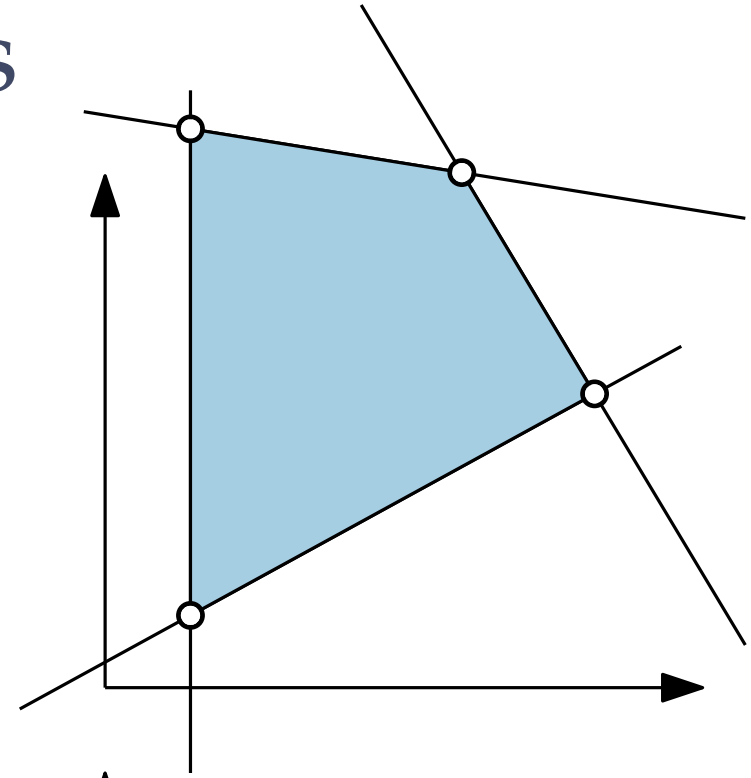


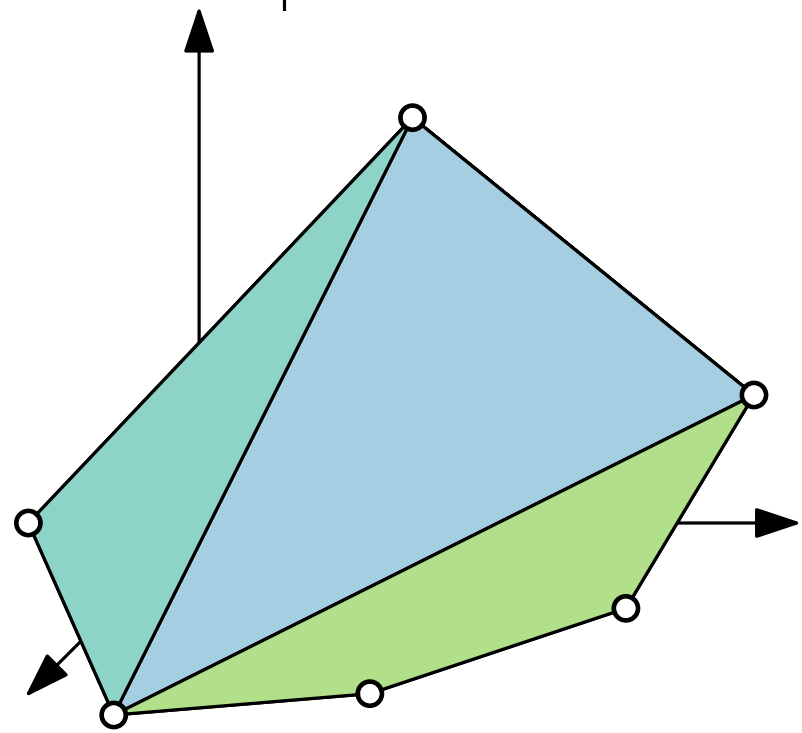


LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

Corners of the polytope are called **extreme point solutions** $\Leftrightarrow$
n linearly independent inequalities (constraints) are satisfied with equality.



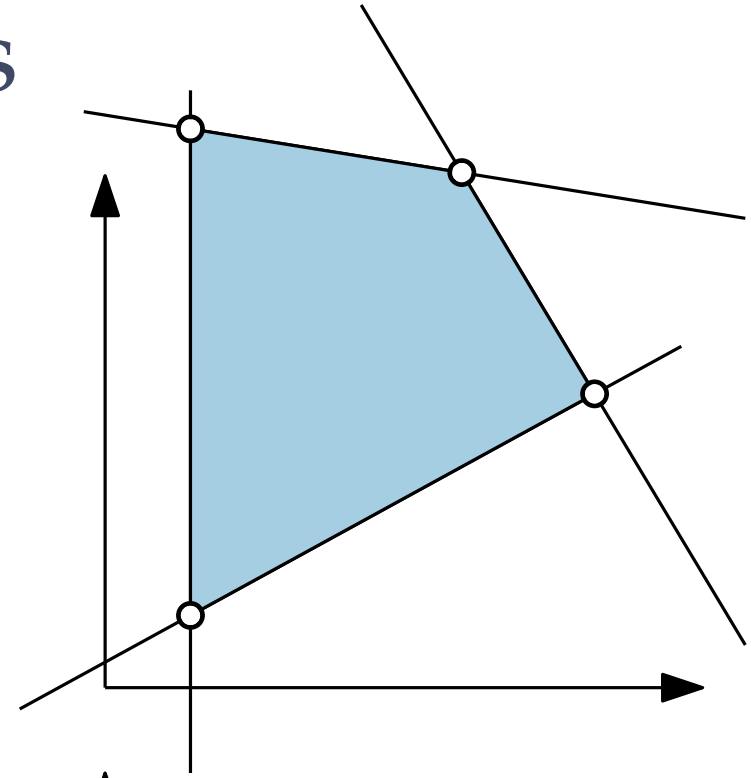


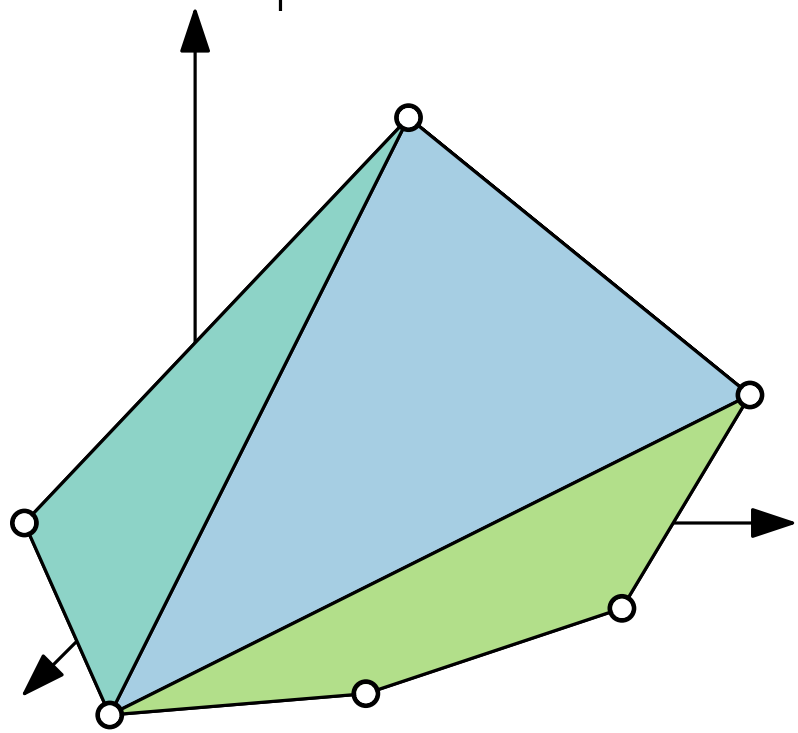
LPs and Convex Polytopes

The feasible solutions of an LP with n variables from a **convex polytope** in $\mathbb{R}^n$ (intersection of halfspaces).

Corners of the polytope are called **extreme point solutions** $\Leftrightarrow$ n linearly independent inequalities (constraints) are satisfied with equality.

When an optimal solution exists, some extreme point will also be optimal.





Integer Linear Programs (ILPs)

$$
\begin{array}{lrcl}
\textbf{minimize} & c^\intercal x & & \\
\textbf{subject to} & Ax & \geq & b \\
& x & \geq & 0
\end{array}
$$

Integer Linear Programs (ILPs)

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \geq & 0 \end{array}$$

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \in & \mathbb{N} \end{array}$$

Integer Linear Programs (ILPs)

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \geq & 0 \end{array}$$

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \in & \mathbb{N} \end{array}$$

Integer Linear Programs (ILPs)

$$\begin{array}{lrcl} \textbf{minimize} & c^\mathsf{T} x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \geq & 0 \end{array}$$

$$\begin{array}{lrcl} \textbf{minimize} & c^\mathsf{T} x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \in & \mathbb{N} \end{array}$$

Many NP-optimization problems can be formulated as ILPs; thus ILPs are NP-hard to solve.

Integer Linear Programs (ILPs)

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \geq & 0 \end{array}$$

$$\begin{array}{lrcl} \textbf{minimize} & c^\top x & & \\ \textbf{subject to} & Ax & \geq & b \\ & x & \in & \mathbb{N} \end{array}$$

Many NP-optimization problems can be formulated as ILPs; thus ILPs are NP-hard to solve.

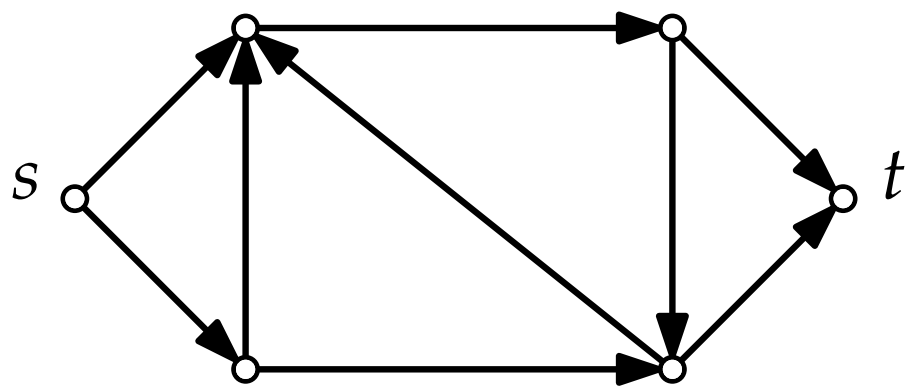
LP-Relaxation provides lower bound: $\text{OPT}_{\text{LP}} \leq \text{OPT}_{\text{ILP}}$

Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

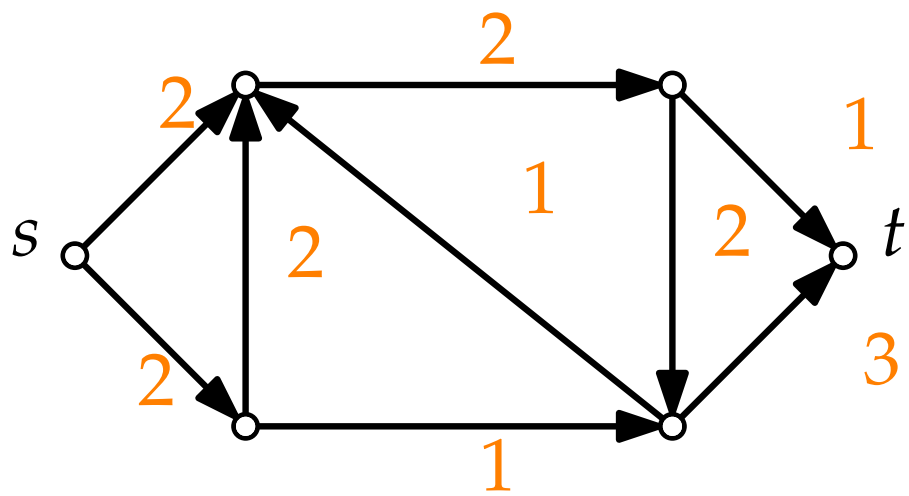
Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.



Max-Flow-Problem

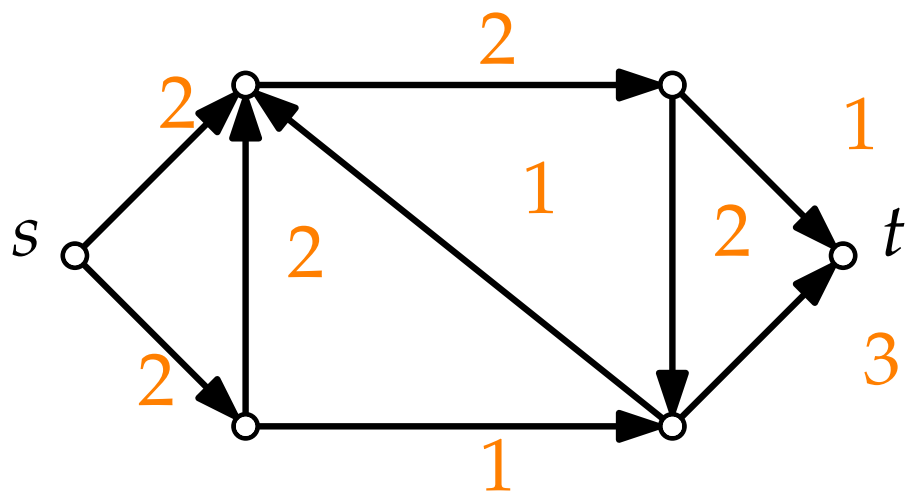
Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.



Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

Find: A maximum s–t-flow (i.e., non-negative edge weights f), such that

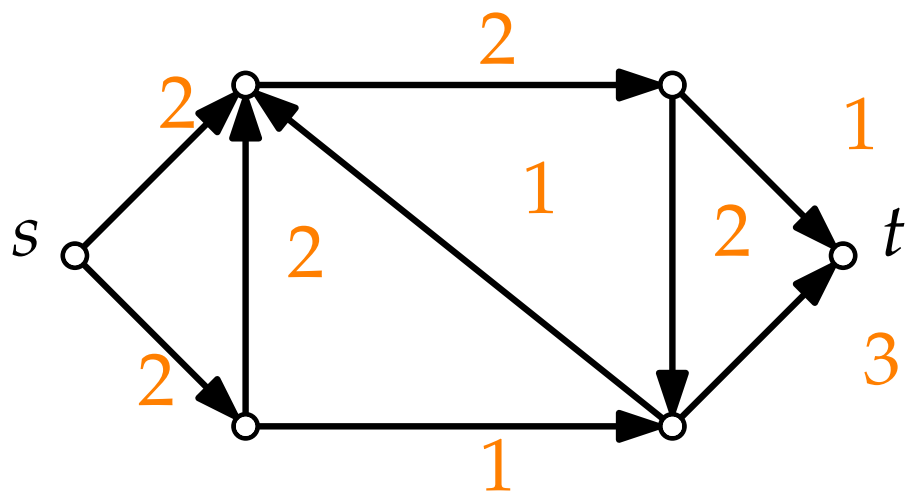


Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

Find: A maximum s–t-flow (i.e., non-negative edge weights f), such that

- $f(u, v) \leq c(u, v)$ for each edge $(u, v) \in E$

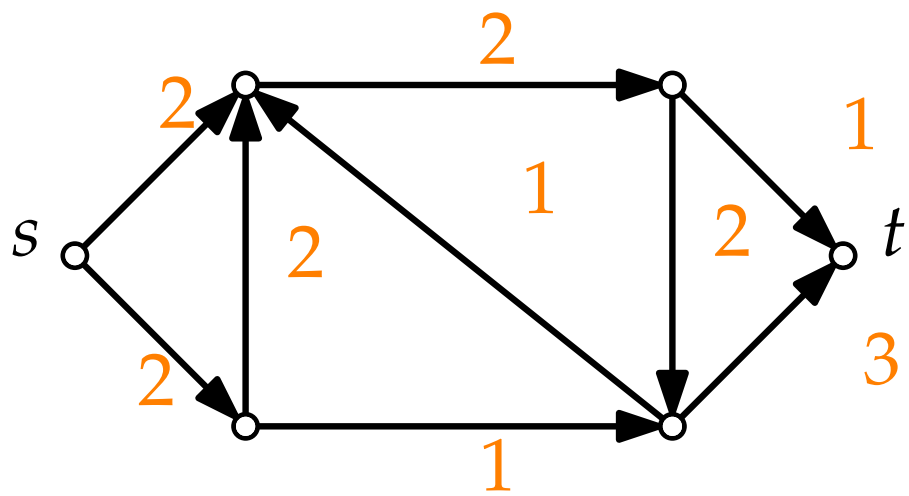


Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

Find: A maximum s–t-flow (i.e., non-negative edge weights f), such that

- $f(u,v) \leq c(u,v)$ for each edge $(u,v) \in E$
- $\sum_{(u,v)\in E} f(u,v) = \sum_{(v,z)\in E} f(v,z)$ for each vertex $v \in V - \{s,t\}$

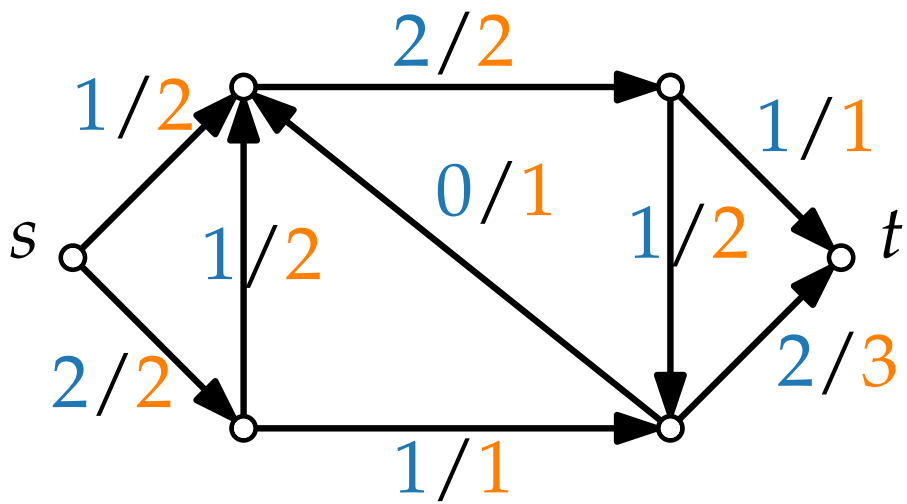


Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

Find: A maximum s–t-flow (i.e., non-negative edge weights f), such that

- $f(u,v) \leq c(u,v)$ for each edge $(u,v) \in E$
- $\sum_{(u,v)\in E} f(u,v) = \sum_{(v,z)\in E} f(v,z)$ for each vertex $v \in V - \{s,t\}$



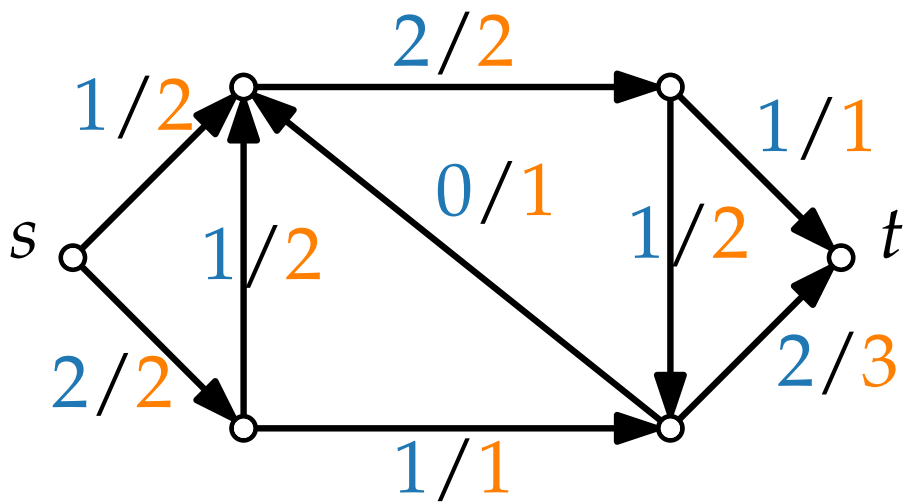
Max-Flow-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

Find: A maximum s–t-flow (i.e., non-negative edge weights f), such that

- $f(u,v) \leq c(u,v)$ for each edge $(u,v) \in E$
- $\sum_{(u,v)\in E} f(u,v) = \sum_{(v,z)\in E} f(v,z)$ for each vertex $v \in V - \{s,t\}$

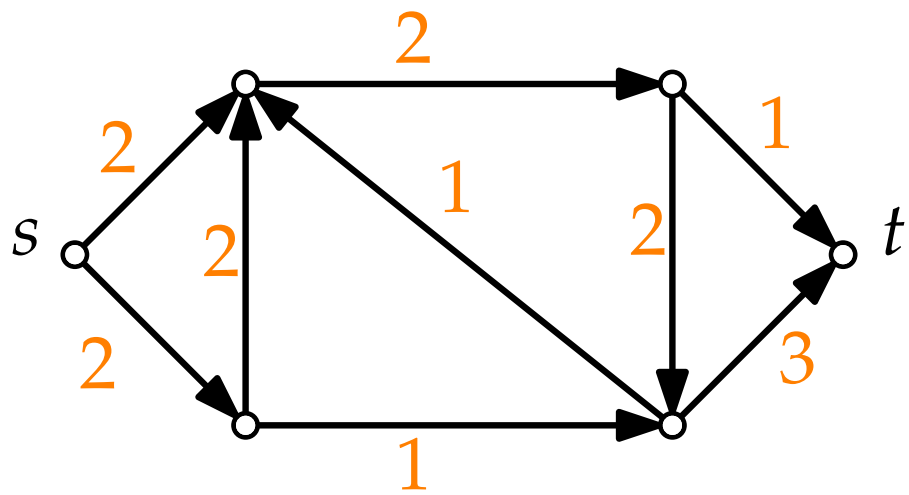
The **flow value** is the inflow to t minus the outflow from t.



Min-Cut-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

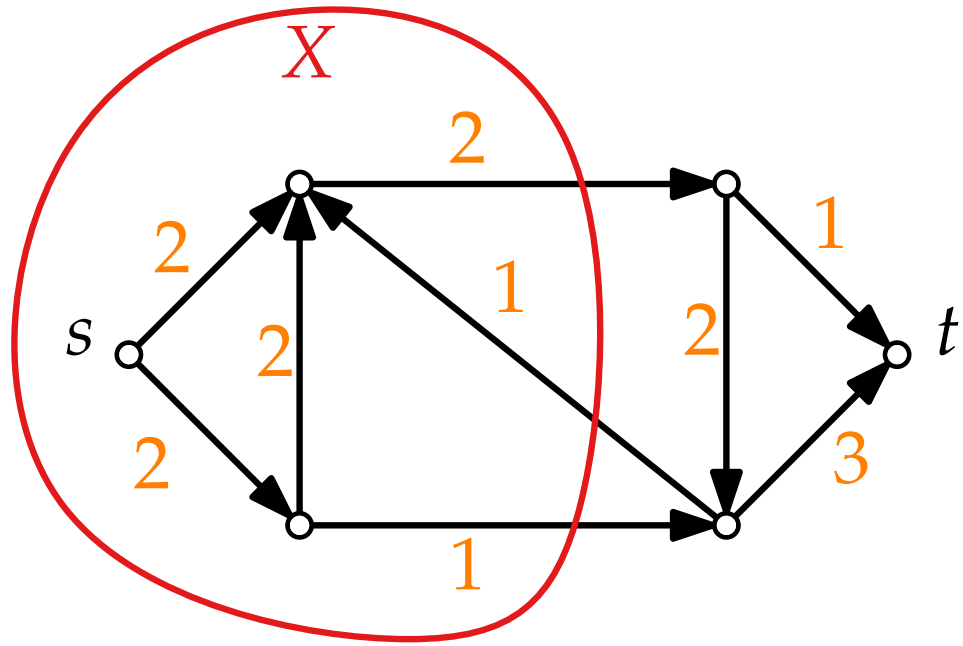
Find: An s–t-cut, i.e., a vertex set X with $s \in X$ and $t \in \overline{X}$, such that the total weight $c(X, \overline{X})$ of the edges from X to $\overline{X}$ is minimum.



Min-Cut-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

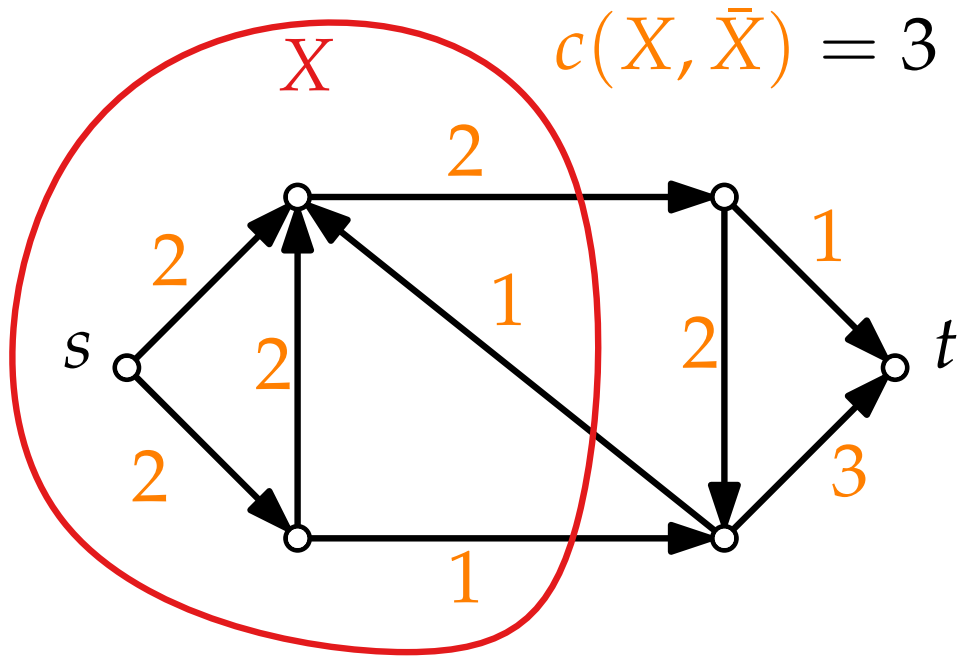
Find: An s–t-cut, i.e., a vertex set X with $s \in X$ and $t \in \overline{X}$, such that the total weight $c(X, \overline{X})$ of the edges from X to $\overline{X}$ is minimum.



Min-Cut-Problem

Given: A directed graph $G = (V, E)$ with edge capacities $c\colon E \to \mathbb{Q}_+$ and two special vertices: the source s and sink t.

Find: An s–t-cut, i.e., a vertex set X with $s \in X$ and $t \in \overline{X}$, such that the total weight $c(X, \overline{X})$ of the edges from X to $\overline{X}$ is minimum.



Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality …

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality …

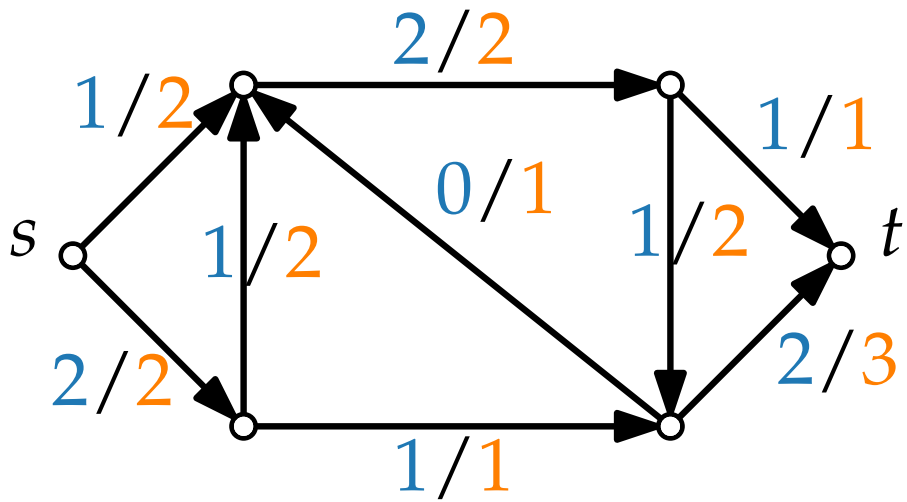
maximize
subject to

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality …

maximize
subject to

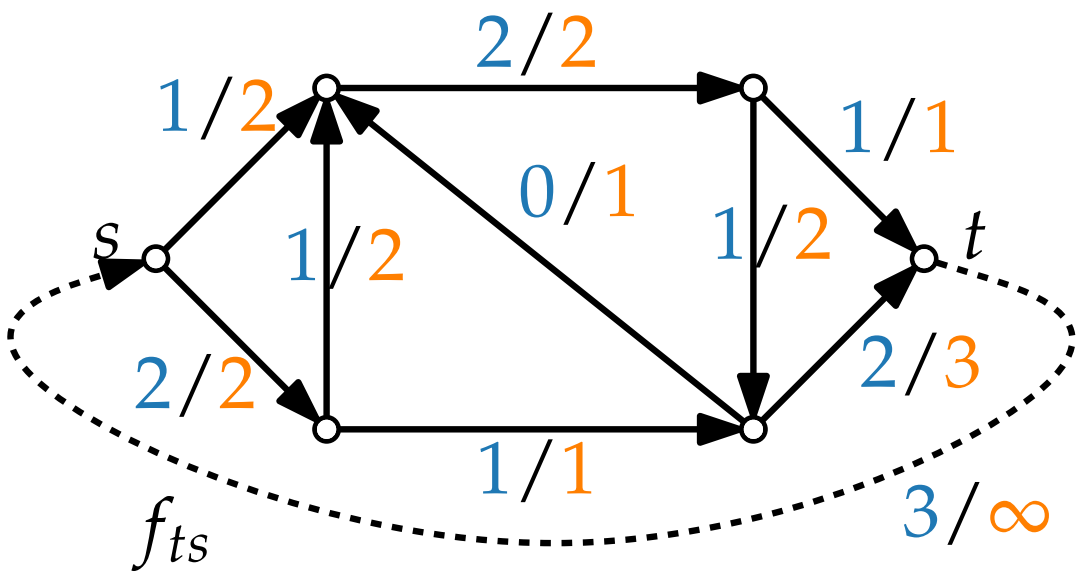


Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize f_{ts}
subject to

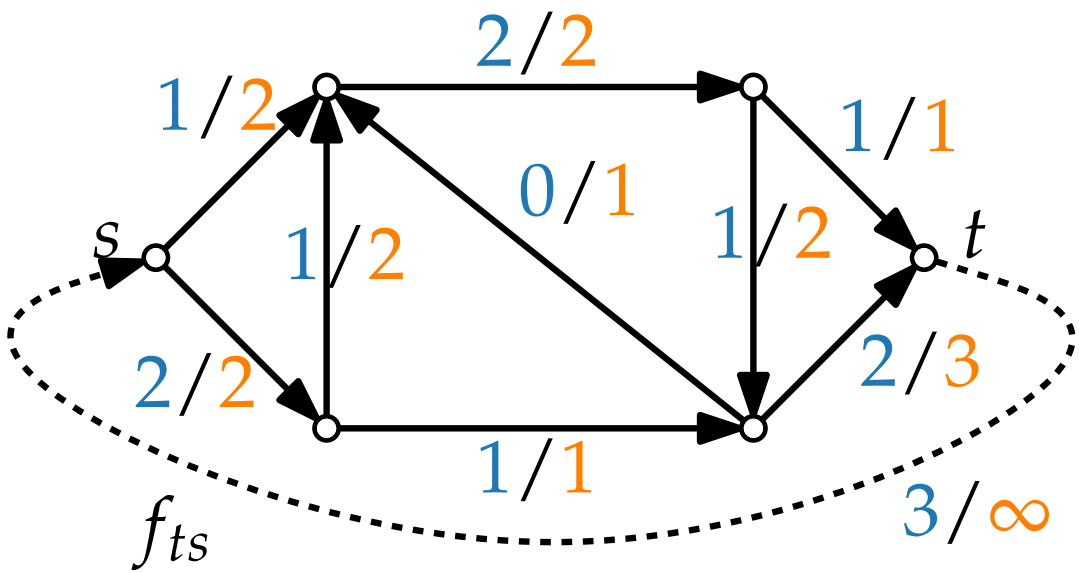


Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize f_{ts}
subject to $f_{uv} \leq c_{uv}$ $\quad \forall (u,v) \in E \setminus \{(t,s)\}$



Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

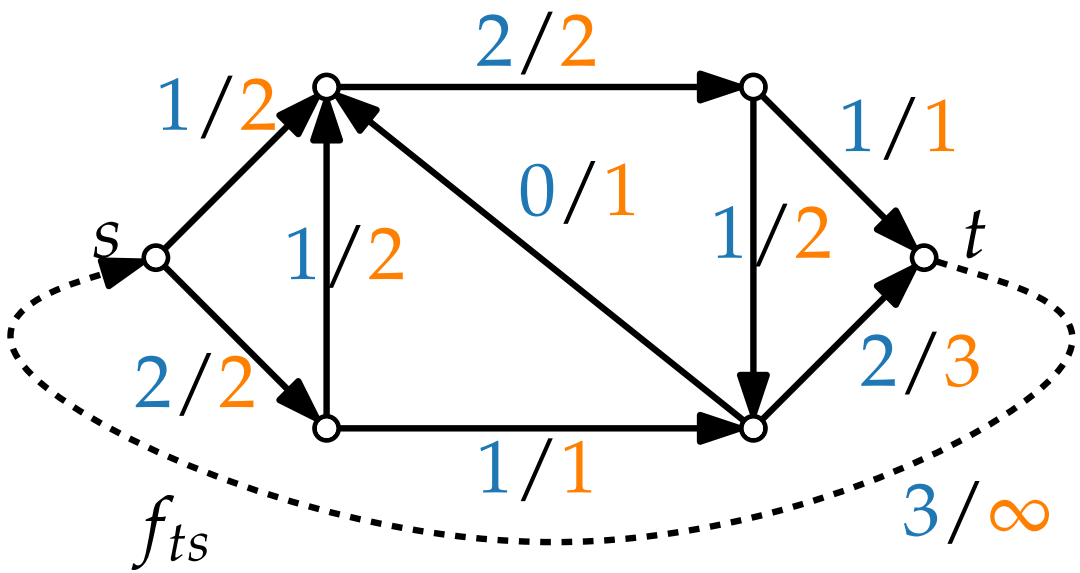
Proof. Special case of LP-Duality ...

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$



Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

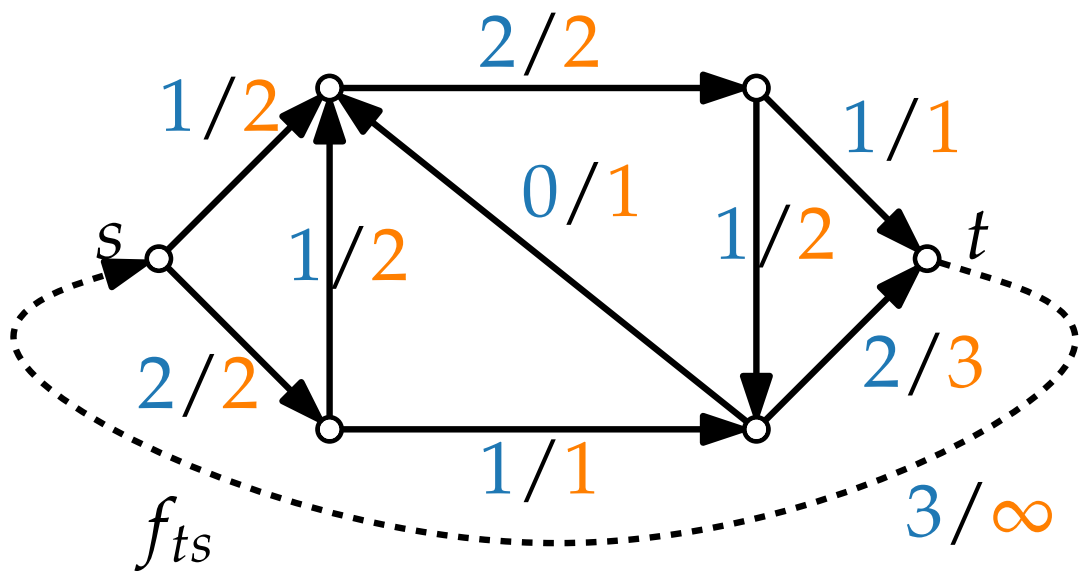
maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$



Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\top x$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv})$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts}$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$?

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}, p_u

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}, p_u, p_v

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}, p_u, p_v

$\Rightarrow d_{uv} - p_u + p_v \geq 0$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}, p_u, p_v

$$\Rightarrow d_{uv} - p_u + p_v \geq 0$$

Which constraints contain f_{ts}?

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}, p_u, p_v

$\Rightarrow d_{uv} - p_u + p_v \geq 0$

Which constraints contain f_{ts}? p_s, p_t

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

maximize $c^\intercal x = \sum_{(u,v)\in E} (0 \cdot f_{uv}) + 1 \cdot f_{ts} \Rightarrow c = (0,\ldots,0,1)$

Which constraints contain $f_{uv} \neq f_{ts}$? d_{uv}, p_u, p_v

$$\Rightarrow d_{uv} - p_u + p_v \geq 0$$

Which constraints contain f_{ts}? p_s, p_t

$$\Rightarrow p_s - p_t \geq 1$$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\} \qquad d_{uv}$$

$$\sum_{u\colon (u,v)\in E} f_{uv} - \sum_{z\colon (v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V \qquad p_v$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

subject to

$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$

Max-Flow-Min-Cut-Theorem

Theorem. The value of a maximum s–t-flow and the weight of a minimum s–t-cut are the same.

Proof. Special case of LP-Duality ...

maximize	f_{ts}		
subject to	$f_{uv} \leq c_{uv}$	$\forall (u,v) \in E \setminus \{(t,s)\}$	d_{uv}
	$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0$	$\forall v \in V$	p_v
	$f_{uv} \geq 0$	$\forall (u,v) \in E$	

minimize	$\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$		
subject to	$d_{uv} - p_u + p_v \geq 0$	$\forall (u,v) \in E \setminus \{(t,s)\}$	
	$p_s - p_t \geq 1$		
	$d_{uv} \geq 0$	$\forall (u,v) \in E$	
	$p_u \geq 0$	$\forall u \in V$	

Dual LP – Interpretation as ILP

$$\begin{aligned}
&\textbf{minimize} && \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv} && \\
&\textbf{subject to} && d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E \setminus \{(t,s)\} \\
& && p_s - p_t \geq 1 && \\
& && d_{uv} \geq 0 && \forall (u,v) \in E \\
& && p_u \geq 0 && \forall u \in V
\end{aligned}$$

Dual LP – Interpretation as ILP

$$\textbf{minimize} \quad \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$$

$$\textbf{subject to} \quad d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \cancel{\geq 0} \in \{0,1\} \qquad \forall (u,v) \in E$$

$$p_u \cancel{\geq 0} \in \{0,1\} \qquad \forall u \in V$$

Dual LP – Interpretation as ILP

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

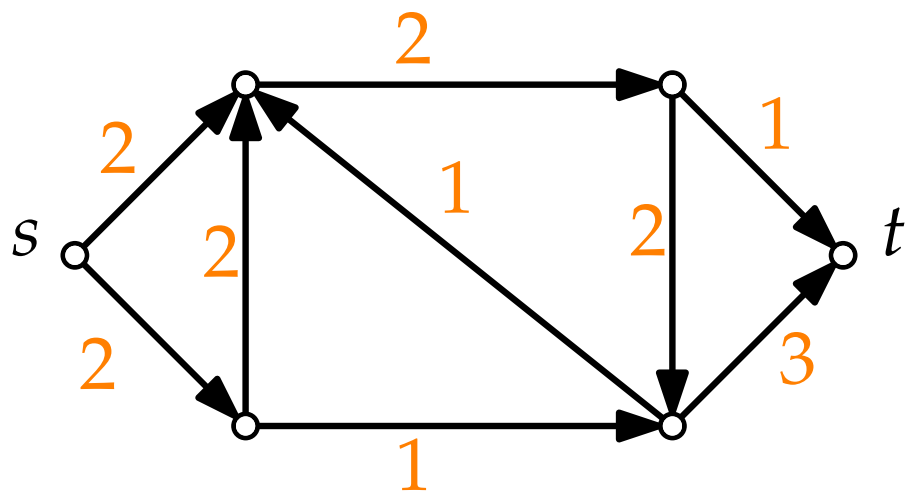
subject to

$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$

$p_s - p_t \geq 1$

$d_{uv} \cancel{\geq 0} \in \{0,1\} \qquad \forall (u,v) \in E$

$p_u \cancel{\geq 0} \in \{0,1\} \qquad \forall u \in V$



Dual LP – Interpretation as ILP

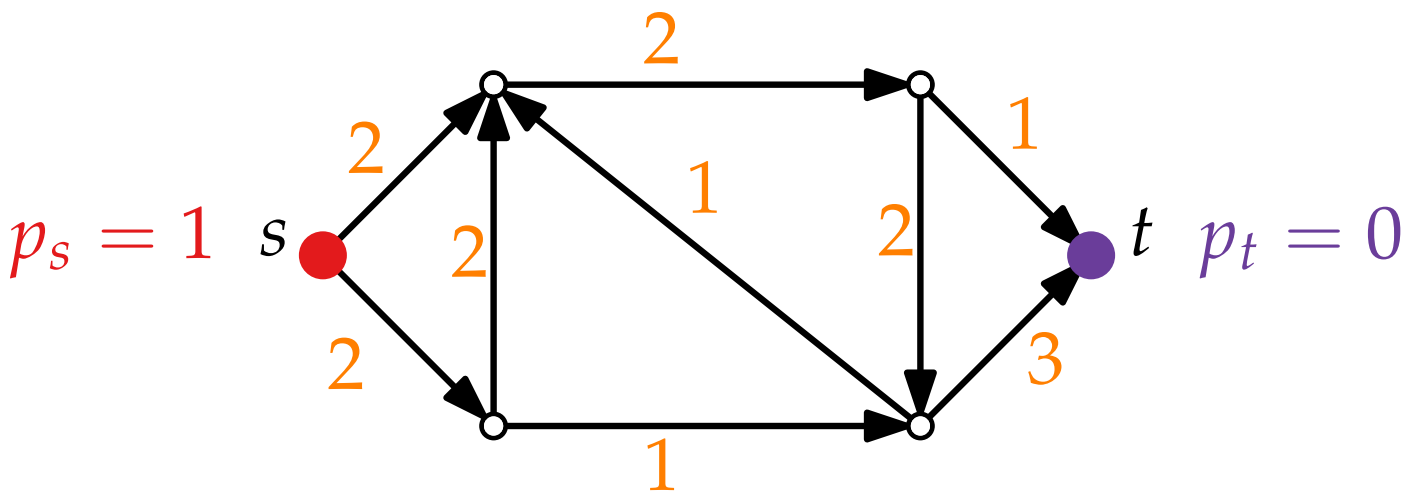
$$\textbf{minimize} \quad \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$$

$$\textbf{subject to} \quad d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \cancel{\geq 0} \in \{0,1\} \qquad \forall (u,v) \in E$$

$$p_u \cancel{\geq 0} \in \{0,1\} \qquad \forall u \in V$$



Dual LP – Interpretation as ILP

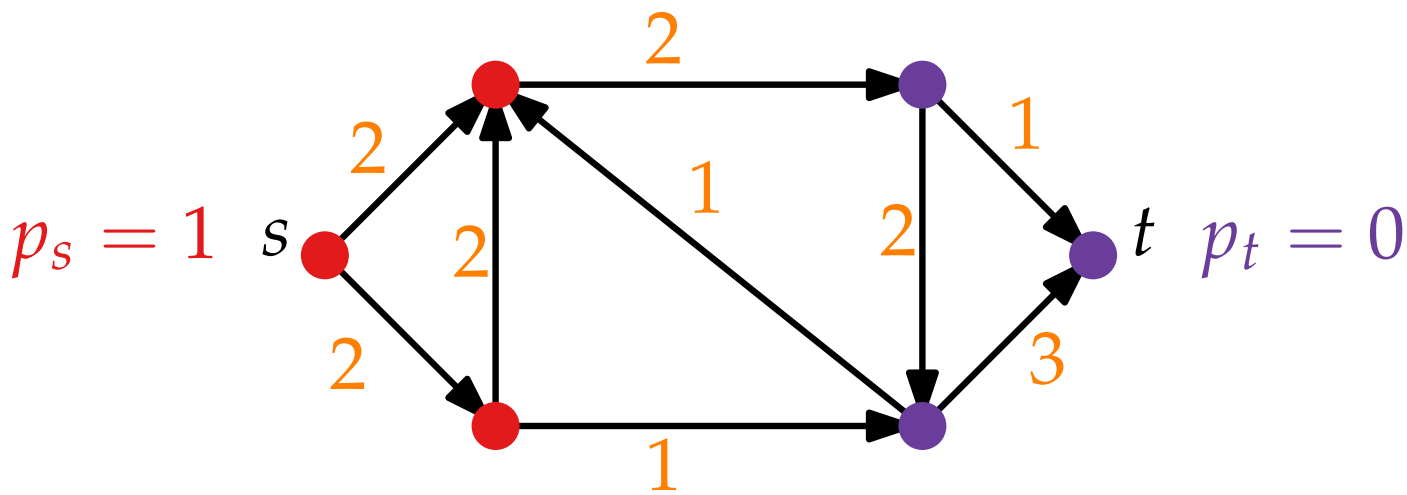
$$\textbf{minimize} \quad \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$$

$$\textbf{subject to} \quad d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \cancel{\geq 0} \in \{0,1\} \qquad \forall (u,v) \in E$$

$$p_u \cancel{\geq 0} \in \{0,1\} \qquad \forall u \in V$$



Dual LP – Interpretation as ILP

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

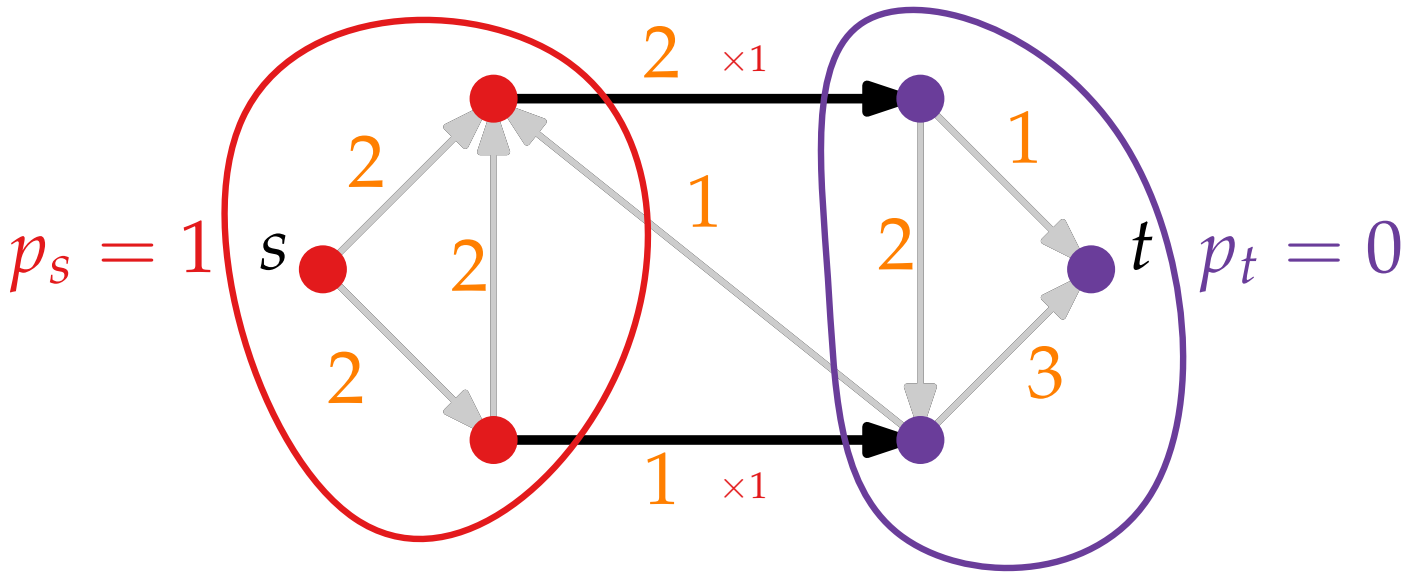
subject to

$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$

$p_s - p_t \geq 1$

$d_{uv} \cancel{\geq 0} \in \{0,1\} \qquad \forall (u,v) \in E$

$p_u \cancel{\geq 0} \in \{0,1\} \qquad \forall u \in V$



Dual LP – Interpretation as ILP

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

subject to

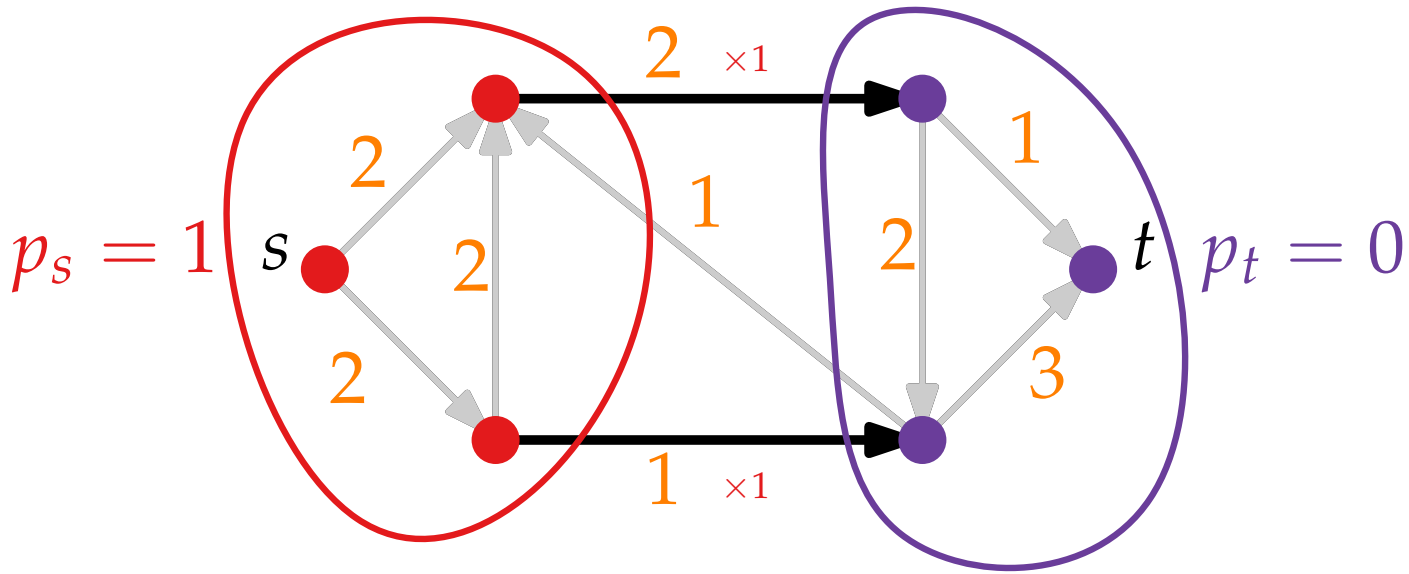
$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$

$p_s - p_t \geq 1$

$d_{uv} \cancel{\geq 0} \in \{0,1\} \qquad \forall (u,v) \in E$

$p_u \cancel{\geq 0} \in \{0,1\} \qquad \forall u \in V$

equivalent to Min-Cut!

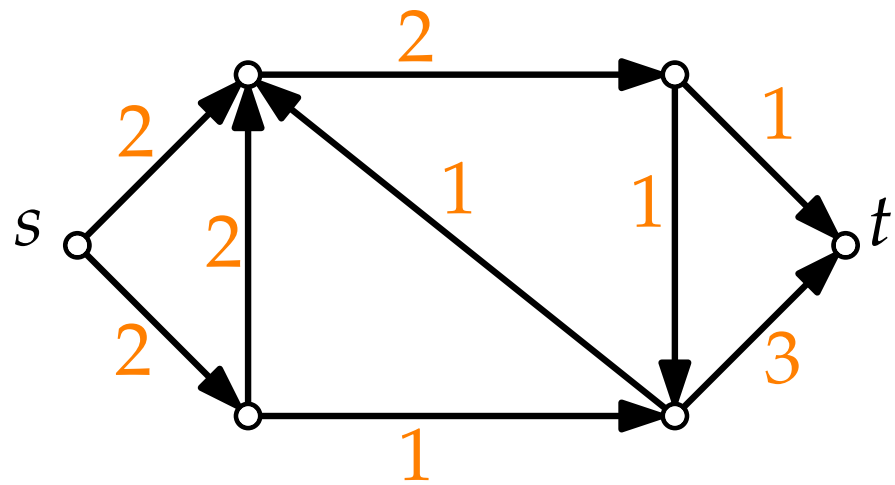


Dual LP – Factional Cuts

$$\begin{aligned}
\textbf{minimize} \quad & \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} \\
\textbf{subject to} \quad & d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E \setminus \{(t,s)\} \\
& p_s - p_t \geq 1 \\
& d_{uv} \geq 0 && \forall (u,v) \in E \\
& p_u \geq 0 && \forall u \in V
\end{aligned}$$

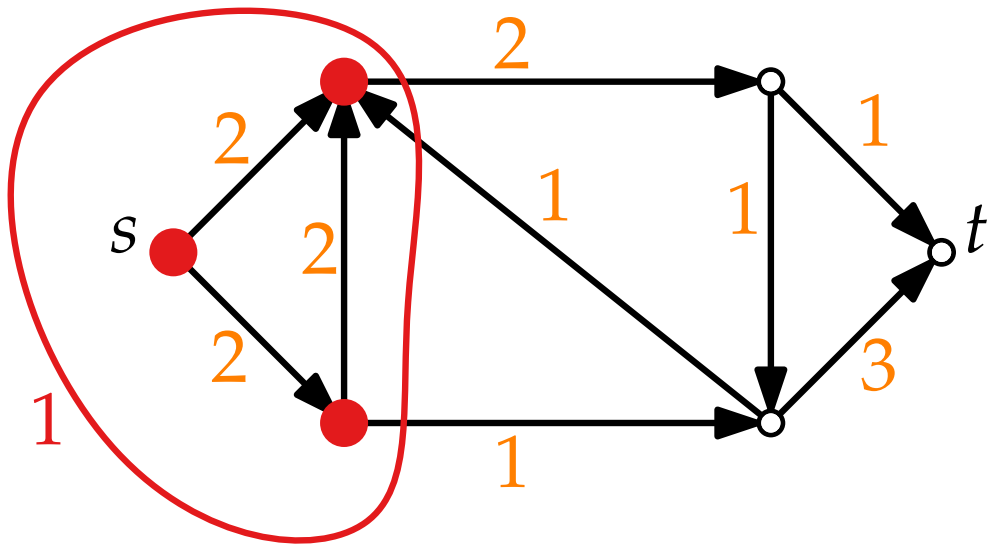
Dual LP – Factional Cuts

$$\begin{aligned}
\textbf{minimize} \quad & \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} \\
\textbf{subject to} \quad & d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E \setminus \{(t,s)\} \\
& p_s - p_t \geq 1 \\
& d_{uv} \geq 0 && \forall (u,v) \in E \\
& p_u \geq 0 && \forall u \in V
\end{aligned}$$



Dual LP – Factional Cuts

$$\begin{aligned}
\textbf{minimize} \quad & \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} \\
\textbf{subject to} \quad & d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E \setminus \{(t,s)\} \\
& p_s - p_t \geq 1 \\
& d_{uv} \geq 0 && \forall (u,v) \in E \\
& p_u \geq 0 && \forall u \in V
\end{aligned}$$



Dual LP – Factional Cuts

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$ ≡ LP-Relaxation of the ILP

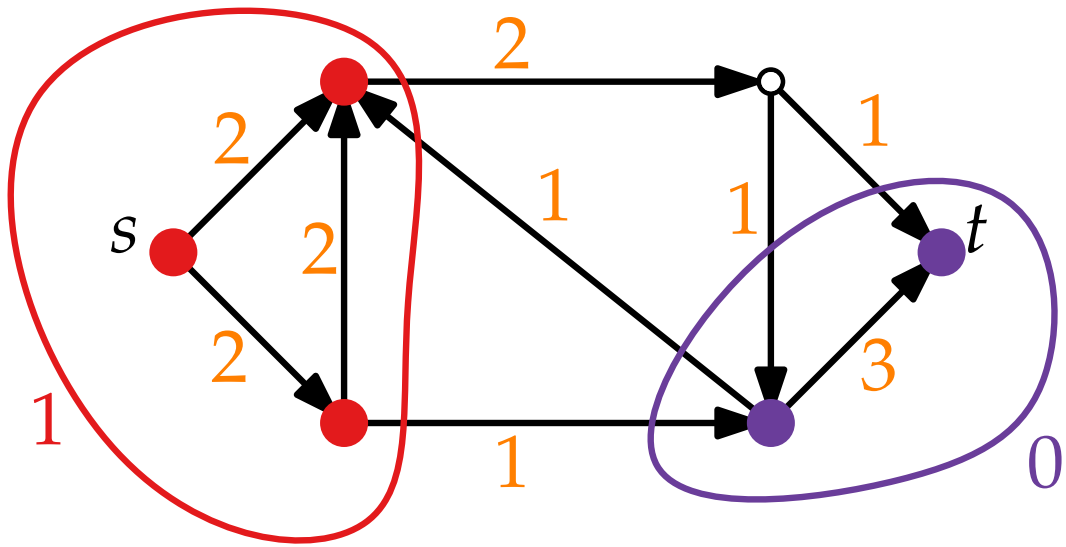
subject to

$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$



Dual LP – Factional Cuts

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$ ≡ LP-Relaxation of the ILP

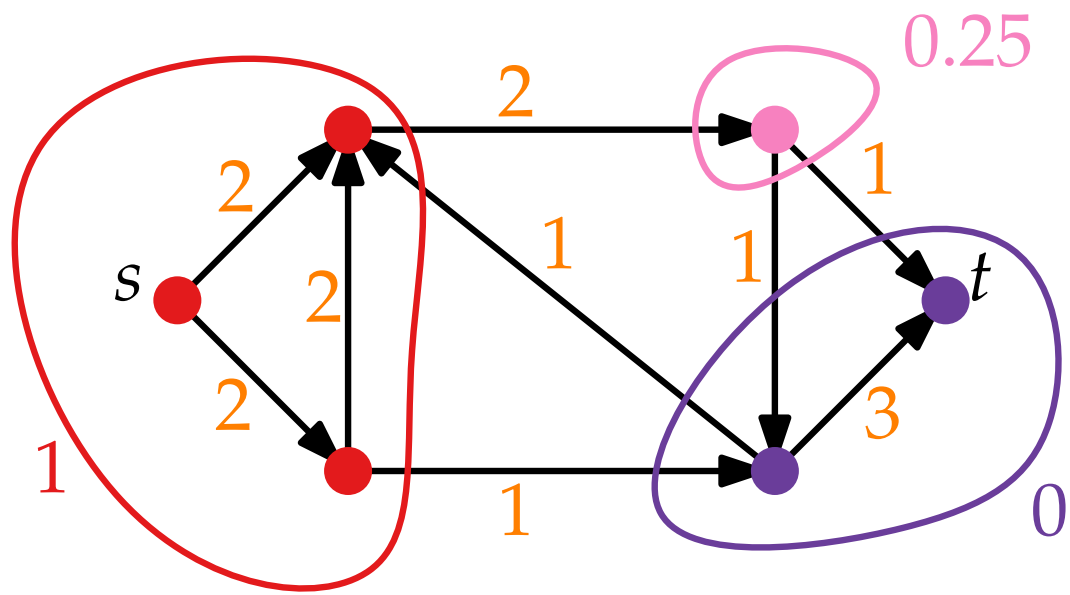
subject to

$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

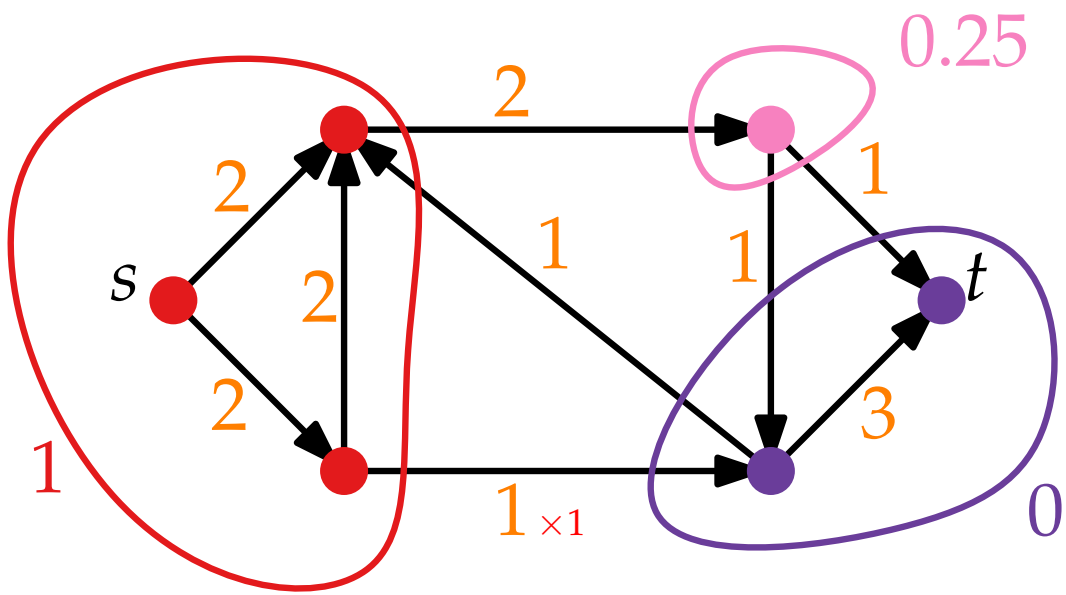
$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$



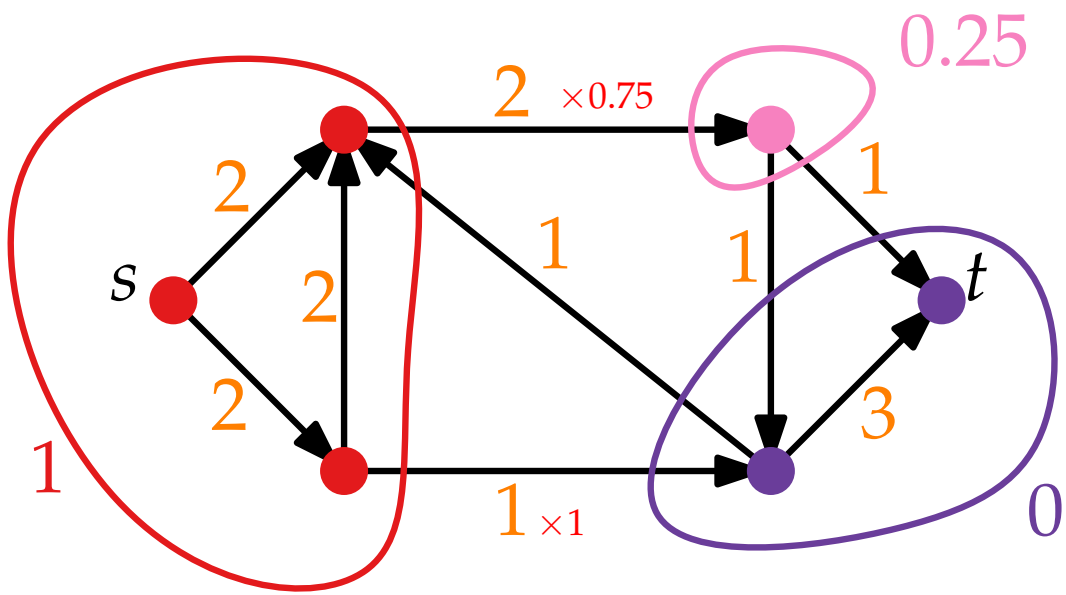
Dual LP – Factional Cuts

$$\begin{aligned}
\textbf{minimize} \quad & \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv}\cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} \\
\textbf{subject to} \quad & d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E\setminus\{(t,s)\} \\
& p_s - p_t \geq 1 \\
& d_{uv} \geq 0 && \forall (u,v) \in E \\
& p_u \geq 0 && \forall u \in V
\end{aligned}$$



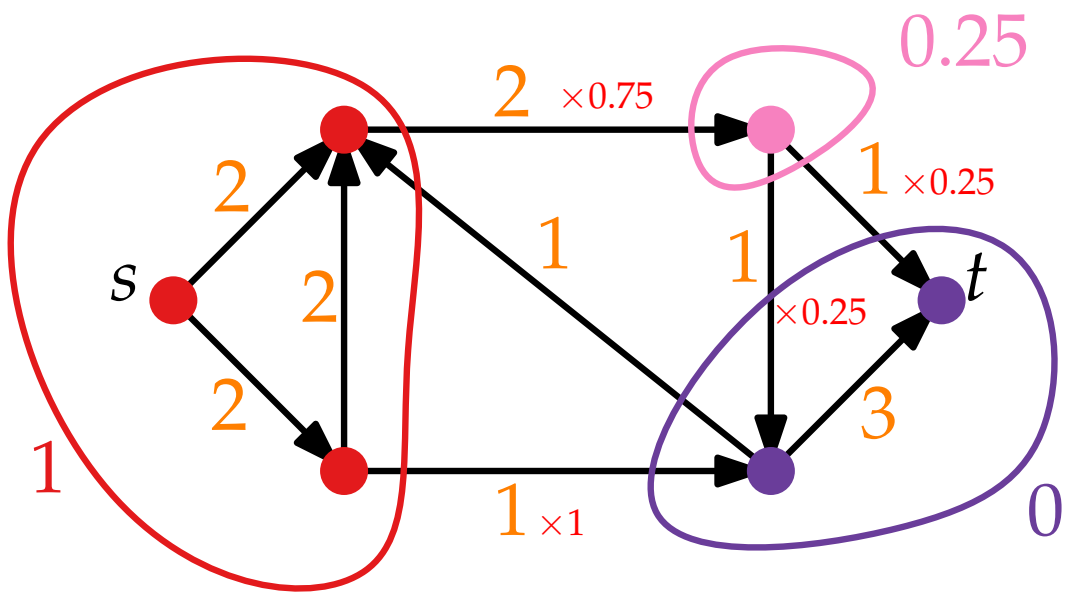
Dual LP – Factional Cuts

$$\begin{aligned}
\textbf{minimize} \quad & \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} \\
\textbf{subject to} \quad & d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E \setminus \{(t,s)\} \\
& p_s - p_t \geq 1 \\
& d_{uv} \geq 0 && \forall (u,v) \in E \\
& p_u \geq 0 && \forall u \in V
\end{aligned}$$



Dual LP – Factional Cuts

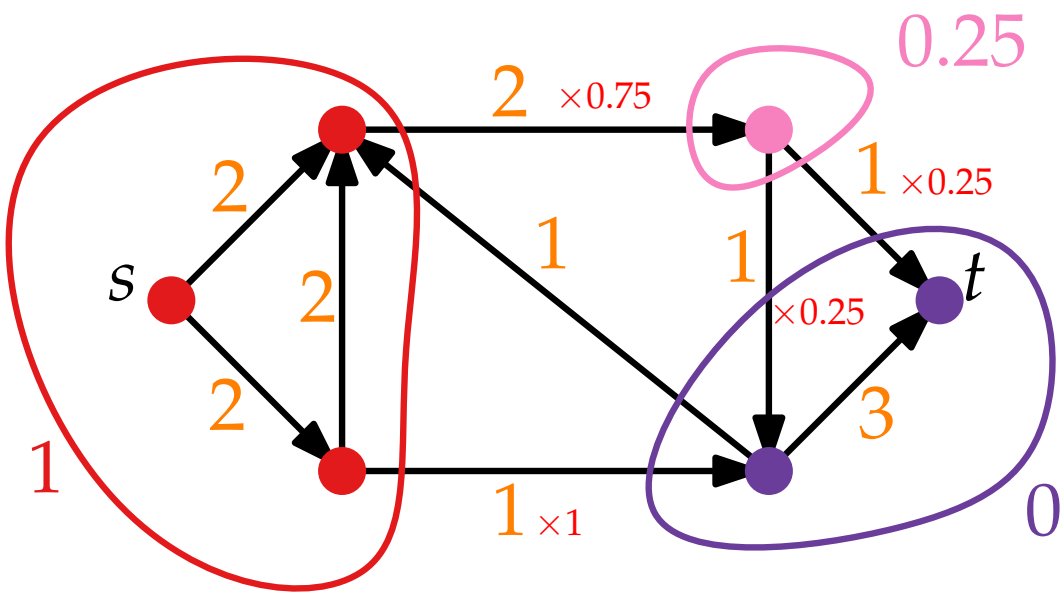
$$
\begin{array}{lll}
\textbf{minimize} & \displaystyle\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv}\cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} & \\
\textbf{subject to} & d_{uv} - p_u + p_v \geq 0 & \forall (u,v) \in E\setminus\{(t,s)\} \\
& p_s - p_t \geq 1 & \\
& d_{uv} \geq 0 & \forall (u,v) \in E \\
& p_u \geq 0 & \forall u \in V
\end{array}
$$



Dual LP – Factional Cuts

$$\begin{aligned}
\textbf{minimize} \quad & \sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv}\cdot d_{uv} \equiv \text{LP-Relaxation of the ILP} \\
\textbf{subject to} \quad & d_{uv} - p_u + p_v \geq 0 && \forall (u,v) \in E\setminus\{(t,s)\} \\
& p_s - p_t \geq 1 \\
& d_{uv} \geq 0 && \forall (u,v) \in E \\
& p_u \geq 0 && \forall u \in V
\end{aligned}$$

Each s–t-path $s = v_0, \ldots, v_k = t$ has length ≥ 1 w.r.t. d:

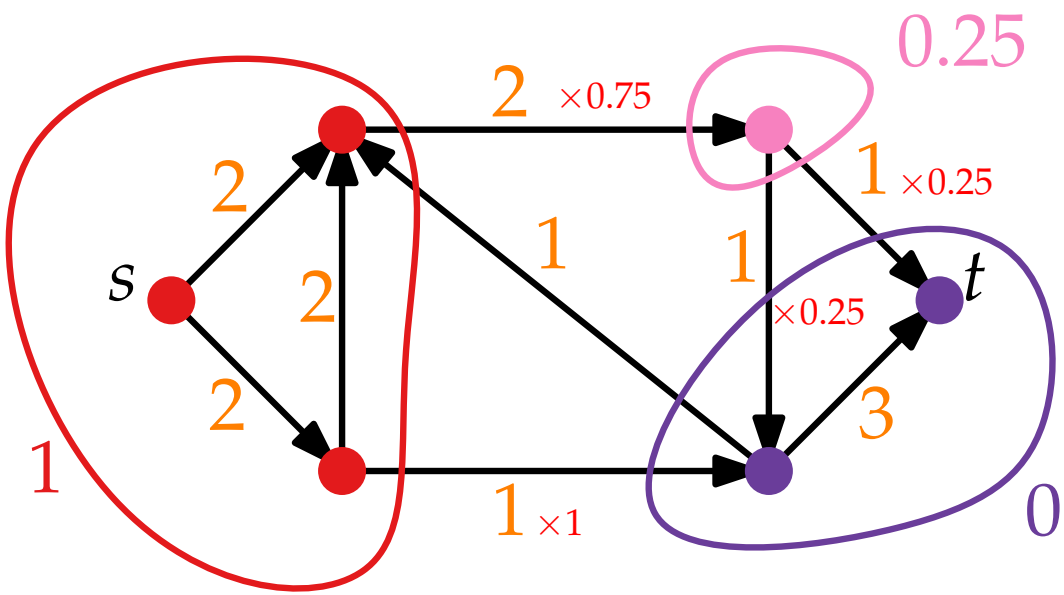


Dual LP – Factional Cuts

minimize	$\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$	$\equiv$ LP-Relaxation of the ILP
subject to	$d_{uv} - p_u + p_v \geq 0$	$\forall (u,v) \in E \setminus \{(t,s)\}$
	$p_s - p_t \geq 1$	
	$d_{uv} \geq 0$	$\forall (u,v) \in E$
	$p_u \geq 0$	$\forall u \in V$

Each s–t-path $s = v_0, \ldots, v_k = t$ has length ≥ 1 w.r.t. d:

$$\sum_{i=0}^{k-1} d_{i,i+1} \geq \sum_{i=0}^{k-1} (p_i - p_{i+1}) = p_s - p_t$$



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

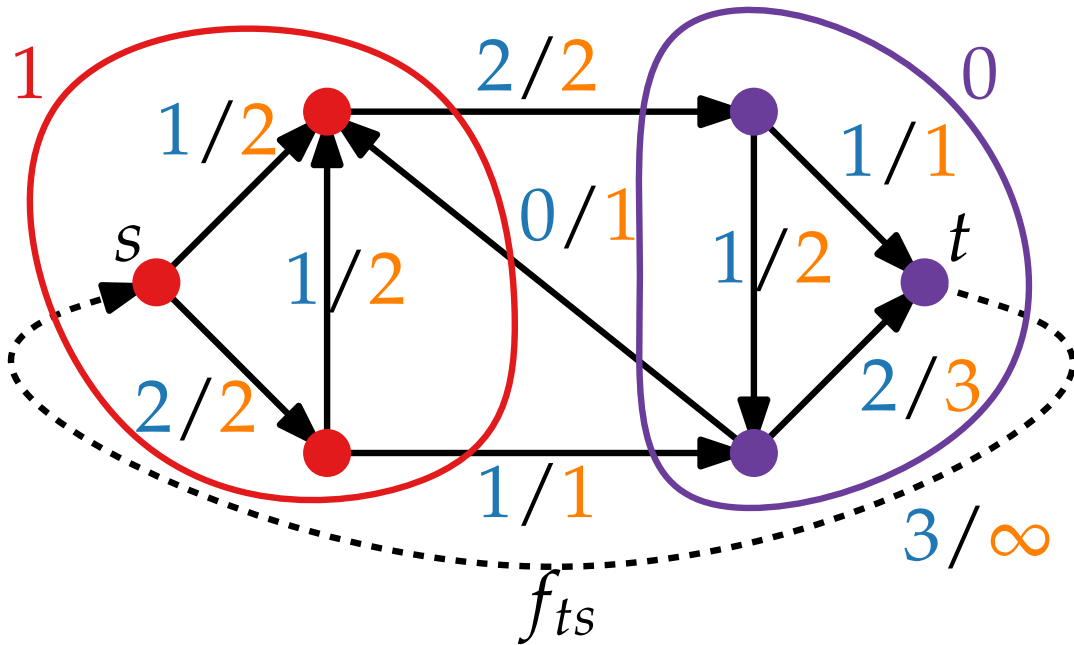
subject to

$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

subject to

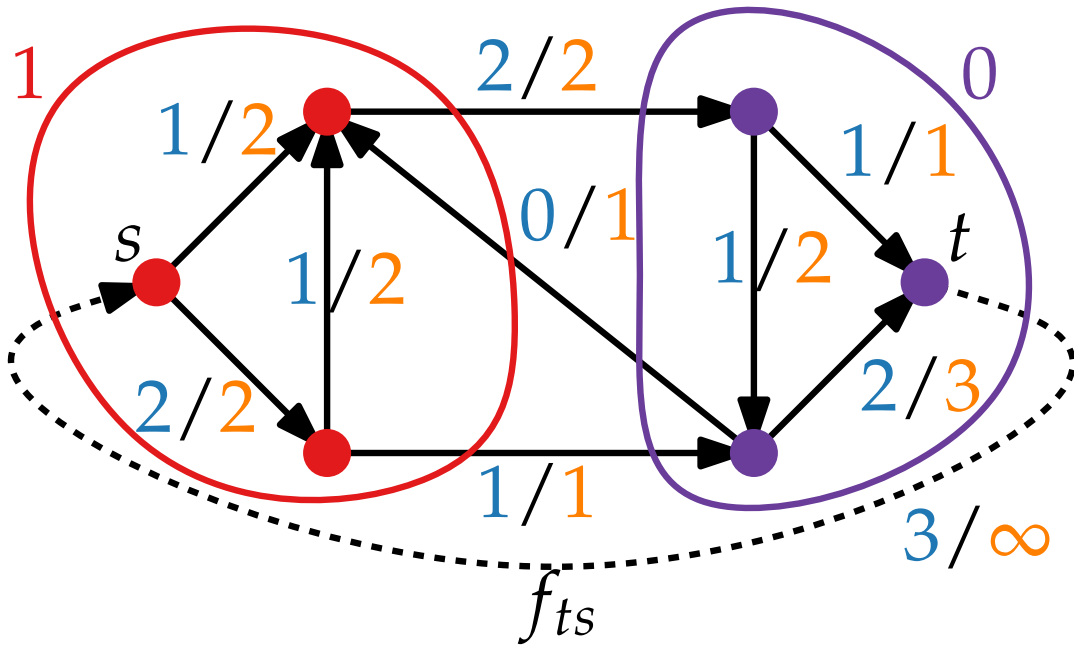
$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$

For a max flow and min cut:



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

subject to

$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

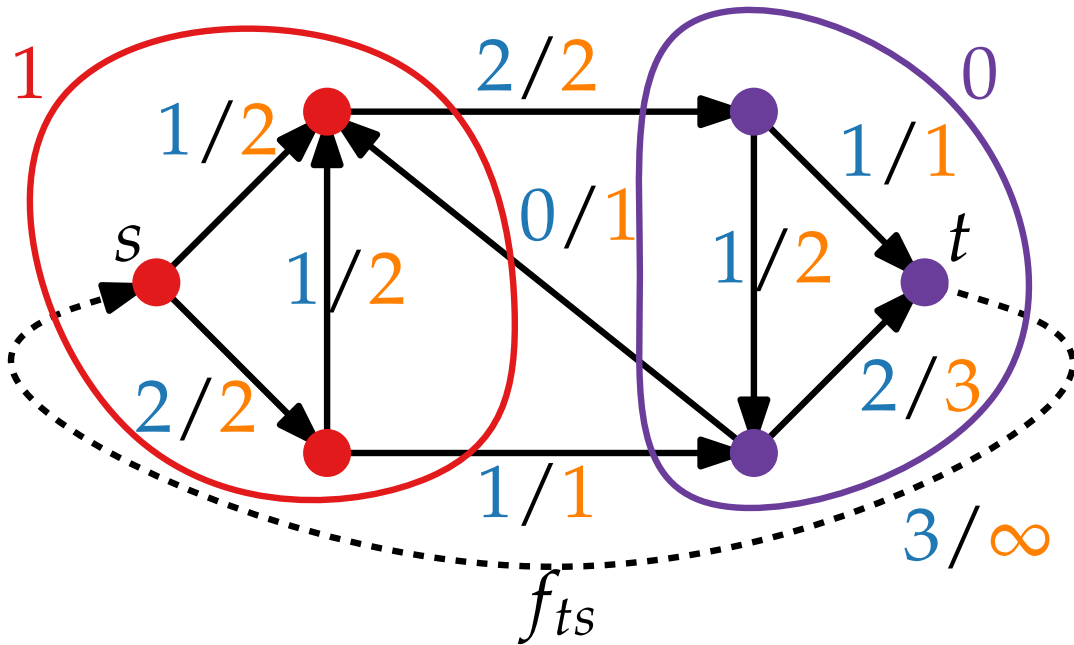
$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$

For a max flow and min cut:

- For each forward edge (u,v) of the cut: $f_{uv} = c_{uv}$.



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

subject to

$$d_{uv} - p_u + p_v \geq 0 \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

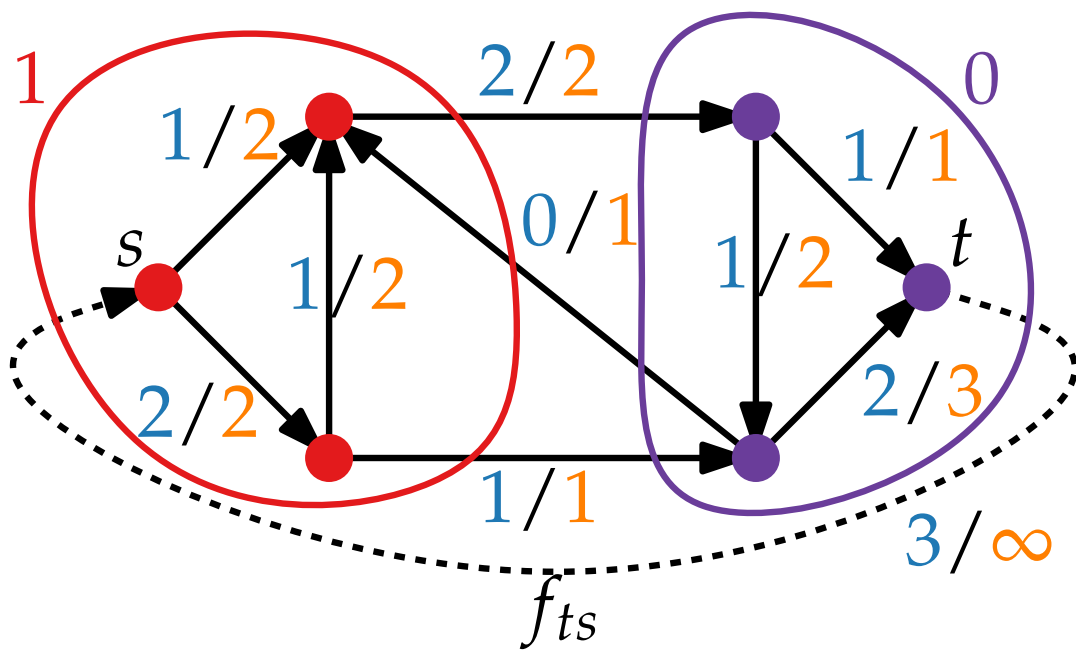
$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0 \qquad \forall (u,v) \in E$$

$$p_u \geq 0 \qquad \forall u \in V$$

For a max flow and min cut:

- For each forward edge (u,v) of the cut: $f_{uv} = c_{uv}$.
  ($d_{uv} = 1$, so by dual CS: $f_{uv} = c_{uv}$.)



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

subject to

$$d_{uv} - p_u + p_v \geq 0$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0$$

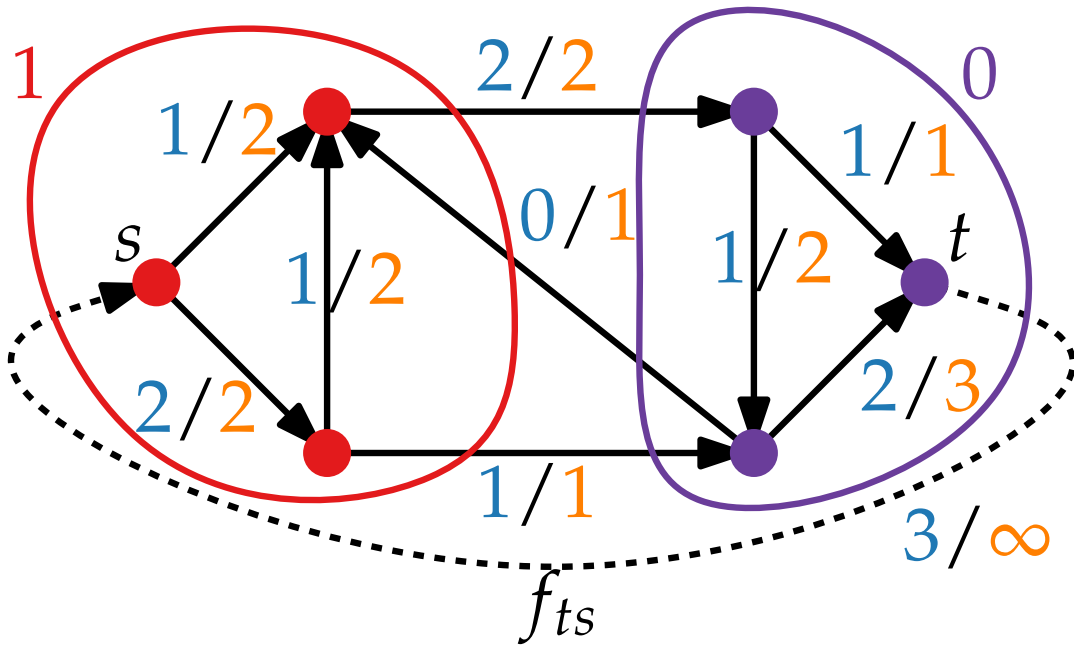
$$p_u \geq 0$$

Primal CS:
$\forall j$: Either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
$\forall i$: Either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

For a max flow and min cut:

- For each forward edge (u,v) of the cut: $f_{uv} = c_{uv}$.
  ($d_{uv} = 1$, so by dual CS: $f_{uv} = c_{uv}$.)



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

subject to

$$d_{uv} - p_u + p_v \geq 0$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0$$

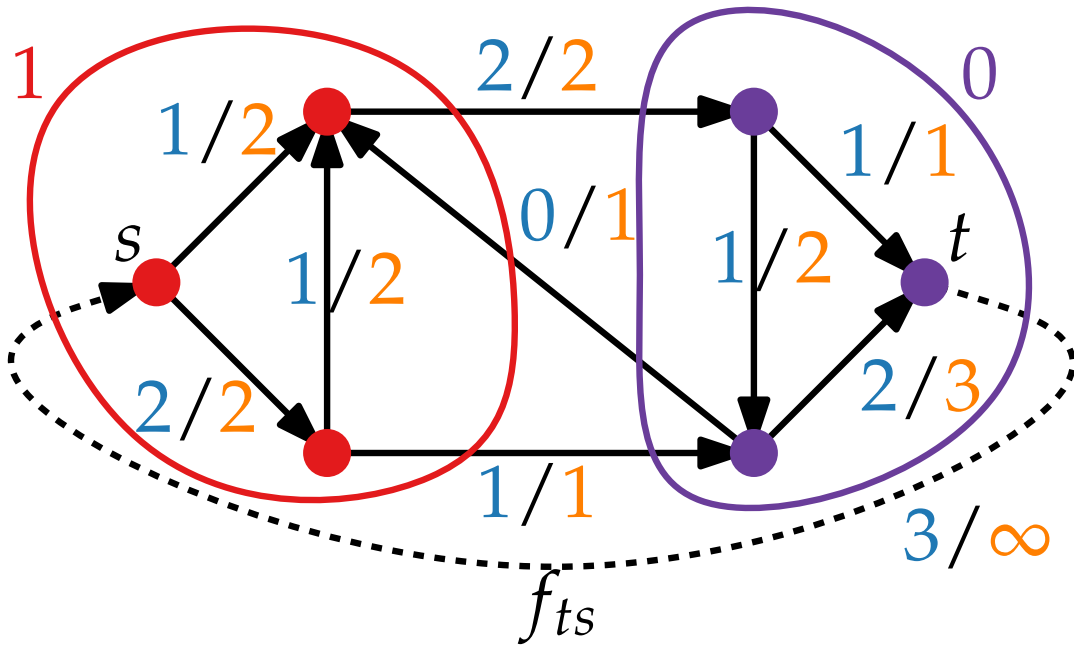
$$p_u \geq 0$$

Primal CS:
$\forall j$: Either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
$\forall i$: Either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

For a max flow and min cut:

- For each forward edge (u,v) of the cut: $f_{uv} = c_{uv}$.
  ($d_{uv} = 1$, so by dual CS: $f_{uv} = c_{uv}$.)
- For each backward edge (u,v) of the cut: $f_{uv} = 0$.



Dual LP – Complementary Slackness

maximize f_{ts}

subject to

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

minimize

$$\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$$

subject to

$$d_{uv} - p_u + p_v \geq 0$$

$$p_s - p_t \geq 1$$

$$d_{uv} \geq 0$$

$$p_u \geq 0$$

Primal CS:
$\forall j$: Either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

Dual CS:
$\forall i$: Either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

For a max flow and min cut:

- For each forward edge (u,v) of the cut: $f_{uv} = c_{uv}$.
  ($d_{uv} = 1$, so by dual CS: $f_{uv} = c_{uv}$.)
- For each backward edge (u,v) of the cut: $f_{uv} = 0$.
  (Otherwise, by primal CS: $d_{uv} - 0 + 1 = 0$.)

